Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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***Abstract*—This paper investigates two-timescale secure transmission for a movable antenna (MA)-enabled multiuser downlink system with a passive eavesdropper (Eve), where the antenna positions are optimized based on statistical channel state information (CSI), and the precoding and artificial noise (AN) are designed based on the instantaneous CSI of the legitimate users (LUs). Specifically, under zero-forcing (ZF) precoding and null-space AN, the antenna positions and the power allocation between the confidential signals and the AN are jointly optimized to maximize the ergodic secrecy sum rate (ESSR) subject to the transmit power and antenna position constraints. Since the ergodic wiretap rate depends on the random ZF directions and AN subspace, we first prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs. Based on this property, a closed-form approximation of the ergodic wiretap rate is derived. This approximation reveals that the MAs can reduce the information leakage and increase the AN received by Eve. Then, an alternating optimization algorithm is developed, where the optimal power allocation for given antenna positions is derived in a secrecy water-filling form. For a given power allocation, each antenna position is updated by a global grid search followed by majorization–minimization refinement, thereby guaranteeing monotonic convergence. Simulation results demonstrate that the proposed design outperforms the benchmark schemes in terms of the ESSR, even for a single LU, while requiring less AN power.**

***Index Terms*—Movable antenna, physical layer security, artificial noise, two-timescale design, statistical CSI, ergodic secrecy rate.**

I. Introduction

PHYSICAL layer security (PLS) exploits the randomness and spatial characteristics of wireless channels to protect confidential messages against eavesdropping, thereby complementing upper-layer cryptography [1]–[3]. In multi-antenna systems, PLS is typically achieved by secure precoding and artificial noise (AN) [4], [5], which steer the confidential signals toward the legitimate users (LUs) and degrade the channel of the eavesdropper (Eve), respectively. However, the effectiveness of both techniques relies on the spatial separability between the LUs and Eve [2]. When Eve is located in a direction close to that of an LU and the line-of-sight (LoS) propagation is dominant, their channels are highly correlated, and a conventional fixed-position antenna (FPA) array can hardly steer the information-bearing beams away from Eve without sacrificing the beamforming gain at the LU. Recently, movable antennas (MAs) [6], [7] and the closely related fluid antennas [8] have emerged as promising technologies to overcome this limitation, where the channel correlation among different receivers is reshaped by adjusting the antenna positions within a given region without additional radio-frequency chains [9]–[11]. For PLS, the transmit beamforming and the antenna positions were jointly optimized in [12] to maximize the secrecy rate with perfect channel state information (CSI), and the case without the instantaneous CSI of the eavesdroppers was considered in [13]. More recently, a deterministic equivalent of the ergodic secrecy rate has been derived in [14] for a multiple-input multiple-output MA system with a multi-antenna Eve and its statistical CSI, and a sensing-assisted secure design has been developed in [15], where the antenna positions enhance both the estimation of the angles of departure (AoDs) of Eve and the secrecy rate. However, the designs in [12]–[14] optimize the antenna positions based on the instantaneous CSI of the legitimate receivers. Under fast fading, such designs require the antennas to be repositioned in each channel coherence interval (CI), thereby incurring nonnegligible movement delay and energy consumption. Moreover, they require the CSI over the entire movable region, whose acquisition entails channel measurements at many antenna positions [16], [17]. On the other hand, the design in [15] assumes LoS channels and thus does not capture the small-scale fading.

To reduce the movement and CSI acquisition overhead, the two-timescale design, which has been widely studied for intelligent reflecting surfaces [18], has been extended to MA systems [16], [17], where the antenna positions are optimized based on the statistical CSI, and the beamforming is designed based on the instantaneous CSI of the resulting effective channels. Specifically, the antenna positions of a multiuser downlink were optimized in [16] to maximize the ergodic sum rate under maximum-ratio transmission and zero-forcing (ZF) precoding, and the multiuser uplink with linear and successive interference cancellation receivers was considered in [17]. However, these works focus on maximizing the ergodic sum rate without considering eavesdropping. To the best of our knowledge, the two-timescale design of MA-enabled secure transmission remains unexplored. Extending the two-timescale MA design to secure transmission is nontrivial for three reasons. First, the ergodic wiretap rate of Eve depends on the random precoding directions and AN subspace induced by the instantaneous CSI of the LUs, whereas the existing ergodic-rate approximations for ZF precoding [16], [17], [19] characterize only the signal power received by the intended users. Second, the antenna positions improve the ergodic rate

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only by decorrelating the users in the rate-oriented design, and thus bring no gain for a single user [16, Remark 2], whereas their impact on the information leakage and the AN received by Eve has not been characterized. Third, the optimal power allocation between the confidential signals and the AN depends on the antenna positions, which determine the information leakage and the AN received by Eve.

To bridge this gap, we jointly optimize the antenna positions and the long-term power allocation of an MA-enabled multiuser downlink with ZF precoding and null-space AN to maximize the ergodic secrecy sum rate (ESSR) based on the statistical CSI of the LUs and Eve. The resulting design requires neither the instantaneous CSI of Eve nor the antenna movement in each CI. First, we prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs. Based on this property, a closed-form approximation of the ergodic wiretap rate is derived. The approximation captures the LoS correlation between Eve and the LUs and is accurate over a wide range of Rician factors. It reveals that, in contrast to the rate-oriented design, the MAs enhance secrecy by reducing the information leakage and increasing the AN received by Eve, even for a single LU. Then, an alternating optimization (AO) algorithm is developed for the resulting approximate ESSR maximization problem. For given antenna positions, the optimal power allocation is derived in semi-closed form. It follows a secrecy water-filling structure and switches off the LUs whose effective channel gains do not exceed that of Eve. For a given power allocation, each antenna position is updated by a global grid search, which exploits a rank-one structure of the objective function, followed by majorization–minimization (MM) refinement. The resulting AO algorithm converges monotonically, and its complexity does not depend on the number of channel realizations due to the closed-form objective function. Finally, simulation results show that the proposed design outperforms the rate-oriented two-timescale MA design [16], antenna selection (AS) from twice as many antennas, and FPA arrays, while requiring considerably less AN power. The secrecy gain persists for a single LU, against an Eve that treats the inter-user interference as noise, under moderate errors in the AoDs of Eve, and when the AoDs of Eve are independent of those of the LUs.

The rest of this paper is organized as follows. Sections II and III present the system model and the approximate ESSR, respectively. Section IV develops the optimization algorithm, Section V provides simulation results, and Section VI concludes the paper.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$ denote the transpose, conjugate, and conjugate transpose, and $|\cdot|$, $\|\cdot\|$, and $\mathrm{tr}(\cdot)$ the modulus, Euclidean norm, and trace, respectively. $[\mathbf{x}]_i$ and $[\mathbf{X}]_{i,j}$ denote the i-th entry of $\mathbf{x}$ and the (i,j)-th entry of $\mathbf{X}$, $\mathrm{diag}(x_1,\ldots,x_M)$ a diagonal matrix, and $\mathbf{X}^{1/2}$ the entrywise square root of a nonnegative diagonal matrix $\mathbf{X}$. $\mathbf{I}_N$ denotes the $N\times N$ identity matrix, $\mathbf{e}_m$ the m-th column of an identity matrix, and $\mathcal{CN}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ the circularly symmetric complex Gaussian distribution. $\mathbf{X}\succeq\mathbf{0}$ ($\mathbf{X}\succ\mathbf{0}$) means that $\mathbf{X}$ is positive semidefinite (definite). $j\triangleq\sqrt{-1}$, $\mathbb{E}\{\cdot\}$ and $\mathrm{Var}\{\cdot\}$ denote the expectation and variance, $\mathrm{Re}\{\cdot\}$ the real part, $[x]^+\triangleq\max\{x,0\}$, and $|\mathcal{S}|$ the cardinality of a set $\mathcal{S}$.

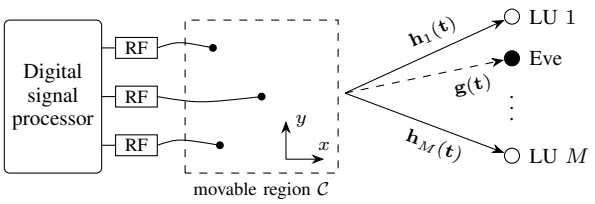


Fig. 1. An MA-enabled multiuser downlink system in the presence of a passive Eve.

II. System Model and Problem Formulation

A. MA-Enabled Secure Multiuser Downlink System

As illustrated in Fig. 1, we consider the downlink of a multiuser system, where a base station (BS) equipped with N MAs serves M single-antenna LUs in the presence of a single-antenna passive Eve. Let $\mathcal{M}\triangleq\{1,\ldots,M\}$ and $\mathcal{N}\triangleq\{1,\ldots,N\}$ denote the index sets of the LUs and the MAs, respectively, where $N\geq M+1$ is required to transmit AN in the null space of the channels of the LUs. The MAs are connected to the RF chains via flexible cables and can move within a two-dimensional (2D) square region $\mathcal{C}\triangleq[-A/2,A/2]\times[-A/2,A/2]$ of side length A [6], [7]. In the local Cartesian coordinate system of $\mathcal{C}$, the position of the n-th MA is denoted by $\mathbf{t}_n\triangleq[x_n,y_n]^T\in\mathcal{C}$, and the positions of all the MAs are collected in $\mathbf{t}\triangleq[\mathbf{t}_1^T,\ldots,\mathbf{t}_N^T]^T\in\mathbb{R}^{2N\times1}$. To avoid the coupling between adjacent antennas, the distance between any two MAs must be no smaller than a minimum distance $D_{\min}$.

Since the size of $\mathcal{C}$ is much smaller than the propagation distance, moving the MAs changes only the phases of the LoS paths, whereas their angles of departure (AoDs) and amplitudes remain unchanged [7]. Let $\theta_m\in[-\pi/2,\pi/2]$ and $\phi_m\in[-\pi/2,\pi/2]$ denote the elevation and azimuth AoDs of the LoS path from the BS to the m-th LU, respectively, and define $\mathbf{a}_m\triangleq[\cos\theta_m\sin\phi_m,\ \sin\theta_m]^T$. Then, the propagation distance of the LoS path from the n-th MA differs from that from the origin of $\mathcal{C}$ by $\mathbf{t}_n^T\mathbf{a}_m$, and the LoS component of the channel from the BS to the m-th LU is given by [16]

$$\bar{\mathbf{h}}_m(\mathbf{t})\triangleq\left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_1^T\mathbf{a}_m},\ldots,e^{j\frac{2\pi}{\lambda}\mathbf{t}_N^T\mathbf{a}_m}\right]^T\in\mathbb{C}^{N\times1},\qquad(1)$$

where λ is the carrier wavelength. Following [16], [17], the channel from the BS to the m-th LU follows Rician fading, i.e.,

$$\mathbf{h}_m(\mathbf{t})=\sqrt{\frac{\kappa_m\beta_m}{\kappa_m+1}}\,\bar{\mathbf{h}}_m(\mathbf{t})+\sqrt{\frac{\beta_m}{\kappa_m+1}}\,\tilde{\mathbf{h}}_m,\qquad(2)$$

where β_m is the large-scale fading coefficient, $\kappa_m\geq0$ is the Rician factor, and $\tilde{\mathbf{h}}_m\sim\mathcal{CN}(\mathbf{0},\mathbf{I}_N)$ is the non-LoS (NLoS)

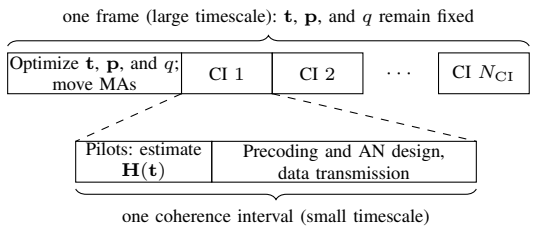


Fig. 2. Two-timescale transmission protocol (CI: coherence interval).

component.[1] Similarly, the channel from the BS to Eve is given by

$$\mathbf{g}(\mathbf{t}) = \sqrt{\frac{\kappa_e \beta_e}{\kappa_e + 1}}\, \bar{\mathbf{g}}(\mathbf{t}) + \sqrt{\frac{\beta_e}{\kappa_e + 1}}\, \tilde{\mathbf{g}}, \tag{3}$$

where β_e, κ_e, and $\tilde{\mathbf{g}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ are defined similarly to those in (2), and $\bar{\mathbf{g}}(\mathbf{t})$ is given by (1) with $\mathbf{a}_m$ replaced by $\mathbf{a}_e \triangleq [\cos\theta_e \sin\phi_e,\ \sin\theta_e]^T$, where θ_e and ϕ_e are the elevation and azimuth AoDs of Eve, respectively. The NLoS components $\tilde{\mathbf{h}}_1, \ldots, \tilde{\mathbf{h}}_M$ and $\tilde{\mathbf{g}}$ are mutually independent. Let $\mathbf{H}(\mathbf{t}) \triangleq [\mathbf{h}_1(\mathbf{t}), \ldots, \mathbf{h}_M(\mathbf{t})] \in \mathbb{C}^{N\times M}$ denote the channel matrix from the BS to the LUs, and let $\bar{\mathbf{H}}(\mathbf{t}) \triangleq [\bar{\mathbf{h}}_1(\mathbf{t}), \ldots, \bar{\mathbf{h}}_M(\mathbf{t})]$ and $\tilde{\mathbf{H}} \triangleq [\tilde{\mathbf{h}}_1, \ldots, \tilde{\mathbf{h}}_M]$ collect its LoS and NLoS components, respectively. Moreover, $\mathbf{K} \triangleq \mathrm{diag}(\kappa_1, \ldots, \kappa_M)$ collects the Rician factors of the LUs. Note that the antenna positions affect the channels only through the LoS components $\bar{\mathbf{H}}(\mathbf{t})$ and $\bar{\mathbf{g}}(\mathbf{t})$.

B. Two-Timescale Transmission Protocol

Different from the instantaneous CSI over the entire region $\mathcal{C}$, the statistical CSI, i.e., $\{\beta_m, \kappa_m, \theta_m, \phi_m\}_{m\in\mathcal{M}}$ of the LUs and $\{\beta_e, \kappa_e, \theta_e, \phi_e\}$ of Eve, is determined by the large-scale fading and the LoS geometry, which vary much more slowly than the small-scale fading. Following [13], [15], we assume that the statistical CSI of Eve is known at the BS.[2]

Accordingly, we adopt the two-timescale protocol in Fig. 2. The time is divided into frames, each of which consists of $N_{\mathrm{CI}} \gg 1$ CIs over which the statistical CSI remains constant. At the beginning of each frame, the BS optimizes the antenna positions $\mathbf{t}$ and the long-term transmission parameters based on the statistical CSI, and moves the MAs accordingly. Then, in each CI, the BS estimates $\mathbf{H}(\mathbf{t})$ via uplink pilots by exploiting the channel reciprocity of time-division duplexing, and designs the precoding and AN. For the purpose of exposition, $\mathbf{H}(\mathbf{t})$ is assumed to be perfectly estimated, whereas $\tilde{\mathbf{g}}$ is never available at the BS. Hence, the MAs are moved only once per frame, and the CSI of the LUs is required only at the current antenna positions.

[1]As in [16], [17], the NLoS components at different antennas are assumed to be independent and identically distributed (i.i.d.), which is reasonable in rich-scattering environments with $D_{\min} \geq \lambda/2$.

[2]The statistical CSI of Eve can be obtained, e.g., from the prior knowledge of the location of Eve, by sensing its direction [15], or from its historical signals if it is an untrusted node.

C. Signal Model and Ergodic Secrecy Rate

In each CI, the BS transmits the confidential symbols $s_m \sim \mathcal{CN}(0,1)$, $m \in \mathcal{M}$, together with AN, i.e.,

$$\mathbf{x} = \sum_{m=1}^{M} \mathbf{w}_m s_m + \mathbf{z}, \tag{4}$$

where $\mathbf{w}_m \in \mathbb{C}^{N\times 1}$ is the precoding vector for the m-th LU, and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$ is the AN with covariance matrix $\mathbf{\Sigma} \succeq \mathbf{0}$, where $s_1, \ldots, s_M$ and $\mathbf{z}$ are mutually independent. The transmit power satisfies $\sum_{m=1}^{M} \|\mathbf{w}_m\|^2 + \mathrm{tr}(\mathbf{\Sigma}) \leq P_{\mathrm{tot}}$, where P_{tot} is the maximum transmit power of the BS. Under the two-timescale protocol, $\mathbf{w}_m$ and $\mathbf{\Sigma}$ are designed based on the instantaneous CSI of the LUs in each CI, and are thus denoted by $\mathbf{w}_m(\mathbf{H}(\mathbf{t}))$ and $\mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))$ to emphasize this dependence.

The signal received at the m-th LU is $y_m = \mathbf{h}_m^H(\mathbf{t})\mathbf{x} + n_m$, where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$ is the additive white Gaussian noise (AWGN). The resulting signal-to-interference-plus-noise ratio (SINR) of the m-th LU is

$$\gamma_m = \frac{|\mathbf{h}_m^H(\mathbf{t})\mathbf{w}_m|^2}{\sum_{i=1, i\neq m}^{M} |\mathbf{h}_m^H(\mathbf{t})\mathbf{w}_i|^2 + \mathbf{h}_m^H(\mathbf{t})\mathbf{\Sigma}\mathbf{h}_m(\mathbf{t}) + \sigma_m^2}. \tag{5}$$

The signal received at Eve is $y_e = \mathbf{g}^H(\mathbf{t})\mathbf{x} + n_e$ with $n_e \sim \mathcal{CN}(0, \sigma_e^2)$. Since the receiver of a passive Eve is unknown to the BS, we consider the worst case where Eve can cancel the signals of the other LUs when intercepting s_m, e.g., via multiuser detection [20], which yields a secrecy guarantee regardless of the decoding capability of Eve. The SINR of Eve for s_m is then

$$\gamma_{e,m} = \frac{|\mathbf{g}^H(\mathbf{t})\mathbf{w}_m|^2}{\mathbf{g}^H(\mathbf{t})\mathbf{\Sigma}\mathbf{g}(\mathbf{t}) + \sigma_e^2}, \tag{6}$$

which upper-bounds that of an Eve treating the signals of the other LUs as interference.

Since the instantaneous CSI of Eve is unknown, the confidential message of each LU is encoded over the CIs of a frame, which experience independent realizations of the NLoS components. By coding over these fading realizations, the following ergodic secrecy rate is achievable for the m-th LU [20], [21]:

$$R_m^{\mathrm{sec}} \triangleq \left[\bar{R}_m - \bar{R}_{e,m}\right]^+, \tag{7}$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1+\gamma_m)\}, \quad \bar{R}_{e,m} \triangleq \mathbb{E}\{\log_2(1+\gamma_{e,m})\} \tag{8}$$

denote the ergodic rate of the m-th LU and the ergodic wiretap rate of Eve for s_m, respectively, and the expectations are taken over $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$ for a given $\mathbf{t}$. The ESSR is defined as $\sum_{m=1}^{M} R_m^{\mathrm{sec}}$.

D. Problem Formulation

Let $\mathcal{W} \triangleq \{\mathbf{w}_1(\mathbf{H}(\mathbf{t})), \ldots, \mathbf{w}_M(\mathbf{H}(\mathbf{t})), \mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))\}$ denote the short-timescale design policy, i.e., the mapping from the instantaneous CSI of the LUs to the precoders and the AN covariance matrix. We aim to maximize the ESSR by jointly

optimizing the long-term antenna positions and the short-timescale policy, which is formulated as

$$\begin{aligned}(\mathrm{P1}):\ \max_{\mathbf{t},\mathcal{W}}\ & \sum_{m=1}^{M} R_m^{\mathrm{sec}} && \text{(9a)}\\ \text{s.t.}\ & \sum_{m=1}^{M} \left\|\mathbf{w}_m(\mathbf{H}(\mathbf{t}))\right\|^2 + \mathrm{tr}\big(\mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))\big) \le P_{\mathrm{tot}}, \\ & \mathbf{\Sigma}(\mathbf{H}(\mathbf{t})) \succeq \mathbf{0},\ \forall \mathbf{H}(\mathbf{t}), && \text{(9b)}\\ & \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, && \text{(9c)}\\ & \|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min},\ \forall n,i \in \mathcal{N},\ n \ne i. && \text{(9d)}\end{aligned}$$

Here, (9b) limits the transmit power in every CI, i.e., for every realization of $\mathbf{H}(\mathbf{t})$, (9c) confines the MAs to the movable region, and (9d) enforces the minimum inter-MA distance.

Problem (P1) is challenging to solve for two reasons. First, the ergodic rates in (8) have no closed-form expressions, and the policy $\mathcal{W}$ needs to be designed for all the realizations of $\mathbf{H}(\mathbf{t})$. Second, since the phases of the LoS components in (1) vary with $\mathbf{t}$, the objective function is nonconcave in $\mathbf{t}$, and the minimum-distance constraints in (9d) are nonconvex. To tackle this problem, we first adopt ZF precoding and null-space AN with a long-term power allocation and derive closed-form approximations of the ergodic rates in Section III, which transform (P1) into a deterministic problem of the long-term variables. Then, an AO algorithm is developed for this problem in Section IV.

III. Two-Timescale ZF-AN Design and Ergodic Secrecy Rate Analysis

A. Short-Timescale ZF Precoding and Null-Space AN

In each CI, the BS adopts ZF precoding with a long-term power allocation based on $\mathbf{H}(\mathbf{t})$, i.e., $\mathbf{w}_m(\mathbf{H}(\mathbf{t})) = \sqrt{p_m}\,\bar{\mathbf{w}}_m$, where $p_m \ge 0$ denotes the power allocated to the m-th LU, and

$$\bar{\mathbf{w}}_m \triangleq \frac{\mathbf{H}(\mathbf{t})\big(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m}{\left\|\mathbf{H}(\mathbf{t})\big(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m\right\|} \tag{10}$$

is the normalized ZF precoding vector of the m-th LU, which satisfies $\mathbf{h}_i^H(\mathbf{t})\bar{\mathbf{w}}_m = 0,\ \forall i \ne m$. Meanwhile, the AN is isotropically distributed in the null space of $\mathbf{H}^H(\mathbf{t})$ [5], [20], i.e., $\mathbf{\Sigma}(\mathbf{H}(\mathbf{t})) = q\,\mathbf{P}^{\perp}(\mathbf{t})$, where $q \ge 0$ denotes the AN power per dimension, and

$$\mathbf{P}^{\perp}(\mathbf{t}) \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{t})\big(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{H}^H(\mathbf{t}) \tag{11}$$

is the orthogonal projection onto the null space of $\mathbf{H}^H(\mathbf{t})$. Since this null space has dimension $N-M$ with probability one, we have $\mathbf{H}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{0}$ and $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N-M$. Hence, the AN causes no interference to the LUs, and the transmit power in each CI is $\sum_{m=1}^{M} p_m + (N-M)q$.

We refer to $\mathbf{p} \triangleq [p_1,\ldots,p_M]^T$ and q as the long-term power allocation, which is optimized based on the statistical CSI and remains fixed within a frame. In contrast, $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are updated in each CI according to $\mathbf{H}(\mathbf{t})$. With this design, the SINRs in (5) and (6) reduce to

$$\gamma_m = \frac{p_m|\mathbf{h}_m^H(\mathbf{t})\bar{\mathbf{w}}_m|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2\big[(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\big]_{m,m}}, \tag{12}$$

$$\gamma_{e,m} = \frac{p_m|\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2}{q\,\mathbf{g}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t}) + \sigma_e^2}, \tag{13}$$

respectively, where the second equality in (12) holds since $\mathbf{h}_m^H(\mathbf{t})\mathbf{H}(\mathbf{t})(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\mathbf{e}_m = 1$ and $\|\mathbf{H}(\mathbf{t})(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\mathbf{e}_m\|^2 = [(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$.

The adopted ZF-AN design requires only the instantaneous CSI of the LUs and eliminates both the inter-user interference and the AN at the LUs. Moreover, since $\mathbf{p}$ and q depend only on the statistical CSI, no optimization problem needs to be solved in each CI. The equal power allocation in [16] and the information–AN power splitting with equal power among the LUs are special cases of this power allocation. With the ZF-AN design, (P1) is transformed into

$$\begin{aligned}(\mathrm{P2}):\ \max_{\mathbf{t},\mathbf{p},q}\ & \sum_{m=1}^{M} R_m^{\mathrm{sec}}(\mathbf{t},p_m,q) && \text{(14a)}\\ \text{s.t.}\ & \sum_{m=1}^{M} p_m + (N-M)q \le P_{\mathrm{tot}}, && \text{(14b)}\\ & p_m \ge 0,\ \forall m \in \mathcal{M}, \quad q \ge 0, && \text{(14c)}\\ & \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, && \text{(14d)}\\ & \|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min},\ \forall n,i \in \mathcal{N},\ n \ne i, && \text{(14e)}\end{aligned}$$

where $R_m^{\mathrm{sec}}(\mathbf{t},p_m,q)$ denotes (7) with the SINRs in (12) and (13), and (14b) guarantees (9b) in every CI. Since the ergodic rates in (14a) still have no closed-form expressions, we derive their approximations in the sequel.

B. Ergodic Rate of the LUs

Define $\mathbf{B} \triangleq \mathrm{diag}\big(\frac{\beta_1}{\kappa_1+1},\ldots,\frac{\beta_M}{\kappa_M+1}\big)$ and $\mathbf{L}(\mathbf{t}) \triangleq \bar{\mathbf{H}}(\mathbf{t})\mathbf{K}^{1/2}$. From (2), the channel matrix of the LUs can be written as

$$\mathbf{H}(\mathbf{t}) = \big(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}\big)\mathbf{B}^{1/2}, \tag{15}$$

where $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ is the normalized channel matrix, whose LoS component is weighted by the square roots of the Rician factors. Accordingly, we define

$$\mathbf{\Phi}(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t})\mathbf{L}(\mathbf{t}) + N\mathbf{I}_M, \qquad \mathbf{\Omega}(\mathbf{t}) \triangleq \mathbf{\Phi}^{-1}(\mathbf{t}), \tag{16}$$

where $\mathbf{\Phi}(\mathbf{t}) = \mathbb{E}\{(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})^H(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})\}$ is the correlation matrix of the normalized channels of the LUs. The ergodic rate of the LUs under ZF precoding has been characterized in [16], [19], which is restated below for completeness.

Lemma 1 ([16], [19]): For given $\mathbf{t}$ and p_m, the ergodic rate of the m-th LU is approximated by

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t},p_m) \triangleq \log_2\big(1 + \eta_m(\mathbf{t})\,p_m\big), \tag{17}$$

where $\eta_m(\mathbf{t}) \triangleq \zeta_m(N-M)/(N\omega_m(\mathbf{t}))$ is the effective channel gain of the m-th LU, $\zeta_m \triangleq \beta_m/((\kappa_m+1)\sigma_m^2)$, and $\omega_m(\mathbf{t}) \triangleq [\mathbf{\Omega}(\mathbf{t})]_{m,m}$.

Lemma 1 follows from Jensen's inequality and a central Wishart approximation of $(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})^H(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})$ with the

same mean $\mathbf{\Phi}(\mathbf{t})$ [19]. It is equivalent to the approximation in [16, Sec. IV-A], reduces to the well-known lower bound $\log_2(1+\zeta_m p_m(N-M))$ for Rayleigh fading [23], and is conservative in the LoS-dominant regime (see Section V-B). Moreover, the antenna positions affect $\tilde{R}_m(\mathbf{t}, p_m)$ only through $\omega_m(\mathbf{t})$. Since $\mathbf{\Phi}(\mathbf{t}) \succeq N\mathbf{I}_M$ and $[\mathbf{\Omega}(\mathbf{t})]_{m,m} \geq 1/[\mathbf{\Phi}(\mathbf{t})]_{m,m} = 1/(N(\kappa_m+1))$ [22], we have

$$\zeta_m(N-M) \leq \eta_m(\mathbf{t}) \leq \zeta_m(N-M)(\kappa_m+1), \tag{18}$$

where the upper bound is achieved if $\bar{\mathbf{h}}_m^H(\mathbf{t})\bar{\mathbf{h}}_i(\mathbf{t}) = 0$ for all $i \neq m$ with $\kappa_i > 0$. Hence, $1/(N\omega_m(\mathbf{t})) \in [1, \kappa_m+1]$ is the LoS gain of the m-th LU that survives the ZF precoding, which can be improved by decorrelating the LoS components of the LUs via the antenna positions.

C. Ergodic Wiretap Rate of Eve

Different from γ_m, the SINR of Eve in (13) depends on the random ZF direction $\bar{\mathbf{w}}_m$ and the random AN subspace, both of which are determined by $\mathbf{H}(\mathbf{t})$. To handle this dependence, we first characterize the information leakage and the AN received by Eve conditioned on $\mathbf{H}(\mathbf{t})$.

Lemma 2: Conditioned on $\mathbf{H}(\mathbf{t})$, the information leakage $X_m \triangleq |\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2$ and the AN power $Y \triangleq \mathbf{g}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t})$ received by Eve are independent, and their conditional means and variances are given by

$$\begin{aligned} \mathbb{E}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e\big(\kappa_e x_m + 1\big), \\ \mathrm{Var}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e^2\big(2\kappa_e x_m + 1\big), \end{aligned} \tag{19}$$

$$\begin{aligned} \mathbb{E}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e\big(\kappa_e y + N - M\big), \\ \mathrm{Var}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e^2\big(2\kappa_e y + N - M\big), \end{aligned} \tag{20}$$

where $\tilde{\beta}_e \triangleq \beta_e/(\kappa_e+1)$, $x_m \triangleq |\bar{\mathbf{g}}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2$, and $y \triangleq \bar{\mathbf{g}}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$.

Proof: Let $\mathbf{U} \triangleq [\mathbf{u}_1, \ldots, \mathbf{u}_{N-M}]$ be an orthonormal basis of the null space of $\mathbf{H}^H(\mathbf{t})$. Then, $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{U}\mathbf{U}^H$ and $Y = \sum_{i=1}^{N-M} |\mathbf{u}_i^H\mathbf{g}(\mathbf{t})|^2$. Moreover, $\bar{\mathbf{w}}_m$ is a unit-norm vector in the column space of $\mathbf{H}(\mathbf{t})$, so that $\mathbf{Q} \triangleq [\bar{\mathbf{w}}_m, \mathbf{U}]$ has orthonormal columns. On the other hand, since $\tilde{\mathbf{g}}$ is independent of $\mathbf{H}(\mathbf{t})$, (3) yields $\mathbf{g}(\mathbf{t})|\mathbf{H}(\mathbf{t}) \sim \mathcal{CN}(\sqrt{\kappa_e\tilde{\beta}_e}\,\bar{\mathbf{g}}(\mathbf{t}), \tilde{\beta}_e\mathbf{I}_N)$. Consequently, $\mathbf{Q}^H\mathbf{g}(\mathbf{t})$ is conditionally Gaussian with covariance matrix $\tilde{\beta}_e\mathbf{Q}^H\mathbf{Q} = \tilde{\beta}_e\mathbf{I}_{N-M+1}$, i.e., its entries $\bar{\mathbf{w}}_m^H\mathbf{g}(\mathbf{t}), \mathbf{u}_1^H\mathbf{g}(\mathbf{t}), \ldots, \mathbf{u}_{N-M}^H\mathbf{g}(\mathbf{t})$ are conditionally independent. Hence, $X_m = |\bar{\mathbf{w}}_m^H\mathbf{g}(\mathbf{t})|^2$ and Y are conditionally independent. Finally, (19) and (20) follow from $\mathbb{E}\{|z|^2\} = |\mu|^2 + \sigma^2$ and $\mathrm{Var}\{|z|^2\} = \sigma^4 + 2|\mu|^2\sigma^2$ for $z \sim \mathcal{CN}(\mu, \sigma^2)$, together with $\sum_{i=1}^{N-M} |\mathbf{u}_i^H\bar{\mathbf{g}}(\mathbf{t})|^2 = y$. ∎

Lemma 2 reveals that, given the channels of the LUs, the information leakage is determined only by the component of the wiretap channel along the ZF direction, whereas the AN received by Eve is determined only by its component in the null space of $\mathbf{H}^H(\mathbf{t})$. Moreover, the conditional relative variances of X_m and Y are $(2\kappa_e x_m+1)/(\kappa_e x_m+1)^2$ and $(2\kappa_e y+N-M)/(\kappa_e y+N-M)^2$, respectively. Hence, X_m concentrates when the LoS leakage $\kappa_e x_m$ is strong, which is the case that dominates the ergodic wiretap rate, and Y concentrates when $N-M$ or $\kappa_e y$ is large. This concentration property motivates the approximation $\mathbb{E}\{\log_2(1+X/Y)\} \approx \log_2(1+\mathbb{E}\{X\}/\mathbb{E}\{Y\})$ [19, Lemma 1], which is accurate when X and Y concentrate around their means and is asymptotically exact as $\kappa_e, \kappa_m \to \infty$, $\forall m \in \mathcal{M}$. Using Lemma 2 and taking the expectation over $\tilde{\mathbf{H}}$, we obtain

$$\bar{R}_{e,m} \approx \log_2\left(1 + \frac{p_m\big(\kappa_e\mathbb{E}\{x_m\}+1\big)}{q\big(\kappa_e\mathbb{E}\{y\}+N-M\big)+\bar{\sigma}_e^2}\right), \tag{21}$$

where $\bar{\sigma}_e^2 \triangleq \sigma_e^2/\tilde{\beta}_e = \sigma_e^2(\kappa_e+1)/\beta_e$ is the normalized noise power of Eve.

In (21), $\mathbb{E}\{x_m\}$ and $\mathbb{E}\{y\}$ are the average LoS energies of Eve along the ZF direction of the m-th LU and in the AN subspace, respectively, which have no closed-form expressions. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of the columns of $\mathbf{H}(\mathbf{t})$, they depend on $\mathbf{H}(\mathbf{t})$ only through the normalized channel matrix $\mathbf{L}(\mathbf{t})+\tilde{\mathbf{H}}$ in (15). We approximate $\mathbb{E}\{x_m\}$ and $\mathbb{E}\{y\}$ by replacing the inverse of the random Gram matrix of $\mathbf{L}(\mathbf{t})+\tilde{\mathbf{H}}$ with the inverse of its mean $\mathbf{\Phi}(\mathbf{t})$, where the NLoS contribution to the AN subspace is calibrated by the identity $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N-M$. These approximations lead to the following result.

Proposition 1: Define $\mathbf{v}(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t}) = \mathbf{K}^{1/2}\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$ and

$$c_m(\mathbf{t}) \triangleq \frac{\big|[\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})]_m\big|^2 + N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}}{\omega_m(\mathbf{t})}, \tag{22}$$

$$d(\mathbf{t}) \triangleq \frac{(N-M)\big(N-\mathbf{v}^H(\mathbf{t})\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})\big)}{N-M+N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t}))}. \tag{23}$$

Then, $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$, $\mathbb{E}\{y\} \approx d(\mathbf{t})$, and the ergodic wiretap rate of Eve for s_m is approximated by

$$\bar{R}_{e,m} \approx \tilde{R}_{e,m}(\mathbf{t}, p_m, q) \triangleq \log_2\left(1 + \frac{\xi_m(\mathbf{t})\,p_m}{q\,\psi(\mathbf{t})+\bar{\sigma}_e^2}\right), \tag{24}$$

where $\xi_m(\mathbf{t}) \triangleq \kappa_e c_m(\mathbf{t}) + 1$ and $\psi(\mathbf{t}) \triangleq \kappa_e d(\mathbf{t}) + N - M$.

Proof: See Appendix A. ∎

The approximations $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{y\} \approx d(\mathbf{t})$ in Proposition 1 are exact for Rayleigh fading and asymptotically exact as the LoS components dominate, and $c_m(\mathbf{t}), d(\mathbf{t}) \in [0, N]$ hold as for their exact counterparts (see Appendix A). Accordingly, $\xi_m(\mathbf{t})$ and $\psi(\mathbf{t})$ represent the normalized information leakage for the m-th LU and the normalized AN power received by Eve, respectively. Different from the existing approximations for ZF precoding, which only involve the diagonal entries of $\mathbf{\Omega}(\mathbf{t})$, Proposition 1 captures the correlation between the LoS components of Eve and the LUs via $\mathbf{v}(\mathbf{t})$.

D. Approximate ESSR

Based on Lemma 1 and Proposition 1, the ergodic secrecy rate of the m-th LU is approximated by

$$R_m^{\mathrm{sec}} \approx \tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q) \triangleq \big[\tilde{R}_m(\mathbf{t}, p_m) - \tilde{R}_{e,m}(\mathbf{t}, p_m, q)\big]^+, \tag{25}$$

which depends only on the statistical CSI and is thus suitable for the large-timescale design. The resulting approximate ESSR reveals how the antenna positions affect the secrecy performance, as discussed below.

1) Secrecy mechanisms of the MAs: From (17), (22), and (23), the antenna positions affect the approximate ESSR

via $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$, i.e., via the LoS array correlations $[\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{H}}(\mathbf{t})]_{i,j} = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_j-\mathbf{a}_i)}$ among the LUs and $[\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})]_i = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_e-\mathbf{a}_i)}$ between Eve and the LUs. A smaller $\omega_m(\mathbf{t})$ yields a larger ZF gain of the m-th LU, which is the only mechanism exploited by the rate-oriented design [16]. In secure transmission, a smaller $c_m(\mathbf{t})$ and a larger $d(\mathbf{t})$ provide two additional mechanisms, i.e., less information leakage and more AN received by Eve. In particular, if $\mathbf{v}(\mathbf{t}) = \mathbf{0}$, i.e., the LoS component of Eve is orthogonal to those of all the LUs, then $c_m(\mathbf{t})$ reduces to its minimum $N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}/\omega_m(\mathbf{t})$ caused by the NLoS components, and $d(\mathbf{t})$ achieves its maximum $N(N-M)/(N-M+N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t})))$ for a given $\mathbf{\Phi}(\mathbf{t})$. Hence, by decorrelating the LoS component of Eve from those of the LUs, the MAs simultaneously reduce the information leakage and increase the AN received by Eve, which is difficult for an FPA array when Eve is located in a direction close to that of an LU. Note that if $\kappa_e = 0$, then $\tilde{R}_{e,m} = \log_2(1 + p_m/(q(N-M)+\bar{\sigma}_e^2))$ does not depend on $\mathbf{t}$, i.e., the statistical CSI of Eve carries no spatial information, and only the first mechanism remains.

2) Single-LU case: If $M = 1$, then $\mathbf{\Phi}(\mathbf{t}) = N(\kappa_1+1)$ is a scalar, and $\eta_1 = \zeta_1(N-1)(\kappa_1+1)$ does not depend on $\mathbf{t}$.[3] In contrast,

$$c_1(\mathbf{t}) = \frac{\kappa_1|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2/N + 1}{\kappa_1+1}, \tag{26}$$

$$d(\mathbf{t}) = \frac{(N-1)(\kappa_1+1)}{(N-1)(\kappa_1+1)+1}\left(N - \frac{\kappa_1|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2}{N(\kappa_1+1)}\right) \tag{27}$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Hence, different from the rate-oriented two-timescale design, where the MAs bring no gain for a single user [16, Remark 2], the MAs enhance the secrecy of a single LU by reducing this correlation.

3) High-SNR behavior and AN saving: Consider the information–AN power splitting with equal power among the LUs, i.e., $p_m = \alpha P_{\mathrm{tot}}/M$ and $q = (1-\alpha)P_{\mathrm{tot}}/(N-M)$, where $\alpha \in (0,1)$ is the power-splitting factor. As $P_{\mathrm{tot}} \to \infty$, we have

$$\tilde{R}_m - \tilde{R}_{e,m} = \log_2\frac{\alpha P_{\mathrm{tot}}\eta_m(\mathbf{t})}{M} - \log_2\left(1+\frac{\alpha\Theta_m(\mathbf{t})}{1-\alpha}\right) + o(1), \tag{28}$$

where $\Theta_m(\mathbf{t}) \triangleq \frac{(N-M)\xi_m(\mathbf{t})}{M\psi(\mathbf{t})}$ is the ratio of the normalized information leakage to the normalized AN power received by Eve. Hence, the high-SNR slope of the ESSR equals M regardless of $\mathbf{t}$, and the antenna positions affect the ESSR through the constant term. Moreover, the value of α that maximizes the sum of the constant terms in (28) is the root of $\sum_{m=1}^{M}\left[\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{\Theta_m-1}{1+\alpha(\Theta_m-1)}\right] = 0$, which, for $\Theta_m = \Theta$, $\forall m \in \mathcal{M}$, is given by

$$\alpha^\star = \frac{1}{1+\sqrt{\Theta}}. \tag{29}$$

Since the MAs can decrease $c_m(\mathbf{t})$ and increase $d(\mathbf{t})$, they decrease Θ and thus increase $\alpha^\star$, i.e., they reduce the AN power required for secure transmission, which is referred to as the AN-saving effect.

[3]For $M = 1$, even the exact ergodic rate $\mathbb{E}\{\log_2(1+p_1\|\mathbf{h}_1(\mathbf{t})\|^2/\sigma_1^2)\}$ does not depend on $\mathbf{t}$, since the distribution of $\|\mathbf{h}_1(\mathbf{t})\|^2$ depends on $\bar{\mathbf{h}}_1(\mathbf{t})$ only through $\|\bar{\mathbf{h}}_1(\mathbf{t})\|^2 = N$.

IV. Proposed Two-Timescale Optimization Algorithm

In this section, we reformulate (P2) based on the approximate ESSR and develop an AO algorithm, which alternately optimizes the power allocation and the antenna positions.

A. Problem Reformulation

By replacing the ergodic secrecy rates in (P2) with their approximations in (25), we consider the following problem, which depends only on the statistical CSI:

$$\begin{aligned}(\mathrm{P3}):\ \max_{\mathbf{t},\mathbf{p},q}\ & F(\mathbf{t},\mathbf{p},q) \triangleq \sum_{m=1}^{M}\left[\log_2\left(1+\eta_m(\mathbf{t})p_m\right)\right. \\ & \left. - \log_2\left(1+\frac{\xi_m(\mathbf{t})p_m}{q\psi(\mathbf{t})+\bar{\sigma}_e^2}\right)\right] \quad (30a)\\ \text{s.t.}\ & \text{(14b)–(14e)},\end{aligned}$$

where the operator $[\cdot]^+$ in (25) is dropped without loss of optimality. Specifically, for any feasible $(\mathbf{t},\mathbf{p},q)$, setting $p_m = 0$ for every LU whose term in (30a) is negative yields another feasible solution, whose objective value equals $\sum_{m=1}^{M}\tilde{R}_m^{\mathrm{sec}}(\mathbf{t},p_m,q)$, since the m-th term depends only on p_m and q for a given $\mathbf{t}$. Hence, (P3) has the same optimal value as its counterpart with $[\cdot]^+$, and every term in (30a) is nonnegative at any optimal solution of (P3). In other words, the per-user power allocation allows the BS to switch off the LUs with nonpositive secrecy rates. In contrast, under equal power allocation among the LUs, the operator $[\cdot]^+$ must be retained.

Problem (P3) is still nonconvex, since F is nonconcave w.r.t. $\mathbf{t}$ and the minimum-distance constraints in (14e) are nonconvex. Moreover, $\mathbf{t}$, $\mathbf{p}$, and q are coupled in (30a), which motivates the AO framework.

B. Power Allocation Optimization for Given $\mathbf{t}$

For given $\mathbf{t}$, the coefficients η_m, ξ_m, and ψ are constants, and (P3) reduces to

$$\begin{aligned}(\mathrm{P3.1}):\ \max_{\mathbf{p},q}\ & \sum_{m=1}^{M} f_m(p_m;q) \quad (31a)\\ \text{s.t.}\ & \text{(14b), (14c)},\end{aligned}$$

where

$$f_m(p;q) \triangleq \log_2(1+\eta_m p) - \log_2\left(1+\eta_{e,m}(q)\,p\right), \tag{32}$$

and $\eta_{e,m}(q) \triangleq \xi_m/(q\psi+\bar{\sigma}_e^2)$ is the effective channel gain of Eve for LU m, which decreases with q. Problem (P3.1) is nonconvex in general, since f_m is not jointly concave in p_m and q. Nevertheless, for a given q, its optimal $\mathbf{p}$ can be obtained in semi-closed form, as shown below. The first-order and second-order derivatives of f_m w.r.t. p are given by

$$\frac{\partial f_m}{\partial p} = \frac{\eta_m - \eta_{e,m}}{\ln 2\,(1+\eta_m p)(1+\eta_{e,m}p)}, \tag{33}$$

$$\frac{\partial^2 f_m}{\partial p^2} = \frac{1}{\ln 2}\left[\frac{\eta_{e,m}^2}{(1+\eta_{e,m}p)^2} - \frac{\eta_m^2}{(1+\eta_m p)^2}\right], \tag{34}$$

where the argument q of $\eta_{e,m}$ is omitted. If $\eta_m \le \eta_{e,m}$, then $f_m(p;q) \le 0$ for all $p \ge 0$, and thus $p_m = 0$ is optimal. If $\eta_m > \eta_{e,m}$, then $\eta_m(1+\eta_{e,m}p) - \eta_{e,m}(1+\eta_m p) = \eta_m - \eta_{e,m} > 0$, i.e., $\frac{\eta_m}{1+\eta_m p} > \frac{\eta_{e,m}}{1+\eta_{e,m}p}$, which implies that f_m is strictly increasing and strictly concave in $p \ge 0$. Let $\mathcal{M}_{\rm a}(q) \triangleq \{m \in \mathcal{M} \,|\, \eta_m > \eta_{e,m}(q)\}$ denote the set of LUs that can achieve positive secrecy rates, and $P_{\rm I}(q) \triangleq P_{\rm tot} - (N-M)q$ the power available for the confidential signals. Then, the optimal $\mathbf{p}$ for a given q is given as follows.

Proposition 2 (Secrecy water-filling): For a given q with $P_{\rm I}(q) > 0$ and $\mathcal{M}_{\rm a}(q) \neq \emptyset$, the optimal power allocation is $p_m^\star = 0$ for $m \notin \mathcal{M}_{\rm a}(q)$ and

$$p_m^\star(\nu) = \left[\frac{2(\rho_m - 1)}{\sqrt{(\eta_m - \eta_{e,m})^2 + 4\eta_m\eta_{e,m}\rho_m} + \eta_m + \eta_{e,m}}\right]^+ \tag{35}$$

for $m \in \mathcal{M}_{\rm a}(q)$, where $\rho_m = \rho_m(\nu) \triangleq (\eta_m - \eta_{e,m})/(\nu \ln 2)$, and the water level $\nu > 0$ is the unique solution of $\sum_{m\in\mathcal{M}_{\rm a}(q)} p_m^\star(\nu) = P_{\rm I}(q)$.

Proof: Since the objective is strictly concave and increasing in $\{p_m\}_{m\in\mathcal{M}_{\rm a}(q)}$ and Slater's condition holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality, and the power constraint is active at the optimum [24]. Let $\nu > 0$ denote the dual variable associated with the power constraint. The KKT conditions yield $\partial f_m/\partial p = \nu$ if $p_m^\star > 0$ and $\partial f_m/\partial p|_{p=0} \le \nu$ otherwise. From (33), the former is equivalent to $(1+\eta_m p)(1+\eta_{e,m}p) = \rho_m(\nu)$, i.e., a quadratic equation in p, whose positive root, written in a numerically stable form, is given by (35). The latter is equivalent to $\rho_m(\nu) \le 1$, in which case (35) gives $p_m^\star = 0$. Finally, since $\sum_{m\in\mathcal{M}_{\rm a}(q)} p_m^\star(\nu)$ is continuous and strictly decreasing in ν over $(0, \max_{m\in\mathcal{M}_{\rm a}(q)}(\eta_m - \eta_{e,m})/\ln 2)$, ranging from $+\infty$ to 0, the water level is unique. ∎

Proposition 2 reveals that (35) reduces to the classic water-filling solution $[1/(\nu\ln 2) - 1/\eta_m]^+$ when $\eta_{e,m} \to 0$. Hence, (35) can be regarded as a secrecy water-filling solution, in which the LUs whose effective channel gains do not exceed that of Eve are switched off. Since $p_m^\star(\nu)$ is nonincreasing in ν, the water level can be found by bisection.

Next, let $G(q) \triangleq \sum_{m\in\mathcal{M}} f_m(p_m^\star; q)$ denote the objective value achieved by the optimal $\mathbf{p}$ for a given q, where $G(q) = 0$ if $\mathcal{M}_{\rm a}(q) = \emptyset$ or $P_{\rm I}(q) = 0$. The optimal AN power per dimension is obtained by solving

$$q^\star = \arg \max_{0\le q\le q_{\max}} G(q), \tag{36}$$

where $q_{\max} \triangleq P_{\rm tot}/(N-M)$. Increasing q reduces the effective channel gains of Eve at the cost of less power for the confidential signals. Since $G(q)$ is not necessarily unimodal, (36) is solved by a coarse grid search followed by a fine grid search around the best grid point, as summarized in Algorithm 1.

C. Antenna Position Optimization for Given $\mathbf{p}$ and q

For given $\mathbf{p}$ and q, (P3) reduces to

$$(\text{P3.2}): \max_{\mathbf{t}} \quad F(\mathbf{t}) \tag{37a}$$

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Input: $\{\eta_m, \xi_m\}_{m\in\mathcal{M}}$, ψ, $\bar{\sigma}_e^2$, $P_{\rm tot}$, and the grid size Q.
Output: $q^\star$ and $\mathbf{p}^\star$.

1: **for** each q in a uniform grid of Q points over $[0, q_{\max}]$ **do**
2: Compute $\eta_{e,m}(q)$ and $\mathcal{M}_{\rm a}(q)$, find ν by bisection such that $\sum_{m\in\mathcal{M}_{\rm a}(q)} p_m^\star(\nu) = P_{\rm I}(q)$ with $p_m^\star(\nu)$ in (35), and evaluate $G(q)$.
3: **end for**
4: Repeat Steps 1–3 over a uniform grid of Q points between the two neighbors of the best grid point, and return the best q and its $\mathbf{p}^\star$.

$$\text{s.t.} \quad \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, \tag{37b}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min},\ \forall n, i \in \mathcal{N},\ n \neq i, \tag{37c}$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$. Problem (P3.2) is challenging for two reasons. First, according to Section III-D, $F(\mathbf{t})$ depends on $\mathbf{t}_n$ through the phases $\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_j - \mathbf{a}_i)$, $i, j \in \mathcal{M}\cup\{e\}$, which rotate by 2π when $\mathbf{t}_n$ moves by $\lambda/\|\mathbf{a}_j - \mathbf{a}_i\|$ along $\mathbf{a}_j - \mathbf{a}_i$. Hence, $F(\mathbf{t})$ is highly oscillatory and has numerous local maxima within $\mathcal{C}$. Second, the minimum-distance constraints are nonconvex. Similar to [16], we optimize the antenna positions in an element-wise manner, i.e., we update $\mathbf{t}_n$ with the other positions fixed, for $n = 1, \ldots, N$ in turn. Let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the other positions fixed at their latest values. The subproblem for the n-th MA is given by

$$(\text{P3.2}.n): \max_{\mathbf{t}_n} \quad F(\mathbf{t}_n) \tag{38a}$$

$$\text{s.t.} \quad \mathbf{t}_n \in \mathcal{C}, \tag{38b}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min},\ \forall i \in \mathcal{N}\setminus\{n\}. \tag{38c}$$

Although (P3.2.n) has only two real variables, $F(\mathbf{t}_n)$ is multimodal. Therefore, a purely local method, e.g., the MM method adopted in [16], converges to the local maximum closest to the current position of the MA, so that the MAs move slowly across $\mathcal{C}$ and the AO iterations may stagnate at an inferior solution (see Section V-C). To overcome this issue, we solve (P3.2.n) by a coarse-to-fine approach: in Stage 1, a global search over a grid of candidate positions locates the best basin of $F(\mathbf{t}_n)$; in Stage 2, the MM method refines the position within this basin. The global search is affordable, since $F(\mathbf{t}_n)$ can be evaluated at any candidate position with only $\mathcal{O}(M^2)$ operations by exploiting the following rank-one structure.

1) Rank-one structure: Let $\mathbf{l}_n(\mathbf{t}_n) \in \mathbb{C}^{M\times 1}$ denote the conjugate transpose of the n-th row of $\mathbf{L}(\mathbf{t})$, i.e.,

$$\mathbf{l}_n(\mathbf{t}_n) = \mathbf{K}^{1/2}\left[e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_1}, \ldots, e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_M}\right]^T. \tag{39}$$

Since $\mathbf{L}^H\mathbf{L} = \sum_{i=1}^N \mathbf{l}_i\mathbf{l}_i^H$ and $\mathbf{L}^H\bar{\mathbf{g}} = \sum_{i=1}^N \mathbf{l}_i e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^T\mathbf{a}_e}$, we have

$$\mathbf{\Phi}(\mathbf{t}) = \mathbf{\Gamma}_n + \mathbf{l}_n(\mathbf{t}_n)\mathbf{l}_n^H(\mathbf{t}_n), \tag{40}$$

$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \mathbf{l}_n(\mathbf{t}_n)e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_e}, \tag{41}$$

where $\mathbf{\Gamma}_n \triangleq \sum_{i\neq n} \mathbf{l}_i(\mathbf{t}_i)\mathbf{l}_i^H(\mathbf{t}_i) + N\mathbf{I}_M$ and $\mathbf{r}_n \triangleq \sum_{i\neq n} \mathbf{l}_i(\mathbf{t}_i)e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^T\mathbf{a}_e}$ do not depend on $\mathbf{t}_n$. By the Sherman–Morrison formula [22],

$$\mathbf{\Omega}(\mathbf{t}) = \mathbf{\Gamma}_n^{-1} - \frac{\mathbf{\Gamma}_n^{-1}\mathbf{l}_n(\mathbf{t}_n)\mathbf{l}_n^H(\mathbf{t}_n)\mathbf{\Gamma}_n^{-1}}{1+\mathbf{l}_n^H(\mathbf{t}_n)\mathbf{\Gamma}_n^{-1}\mathbf{l}_n(\mathbf{t}_n)}. \tag{42}$$

Hence, for given $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, the quantities ω_m, c_m, d, and thus $F(\mathbf{t}_n)$ can be evaluated at any candidate position with only $\mathcal{O}(M^2)$ operations.

2) Stage 1 (global search): Let $\mathcal{G} \subset \mathcal{C}$ denote a uniform grid over $\mathcal{C}$ with spacing d_{g}, and $\mathcal{G}_n \triangleq \{\mathbf{t} \in \mathcal{G} \,|\, \|\mathbf{t}-\mathbf{t}_i\| \geq D_{\min},\ \forall i \neq n\}$ the set of grid points that are feasible for (P3.2.n). Based on (42), we evaluate $F(\mathbf{t}_n)$ at all the points in $\mathcal{G}_n \cup \{\mathbf{t}_n\}$, where $\mathbf{t}_n$ is the current position of the n-th MA, and obtain

$$\hat{\mathbf{t}}_n = \arg\max_{\mathbf{t}\in\mathcal{G}_n\cup\{\mathbf{t}_n\}} F(\mathbf{t}), \tag{43}$$

which requires $\mathcal{O}(|\mathcal{G}|M^2)$ operations, e.g., $|\mathcal{G}| = 1681$ candidate positions for $A = 4\lambda$ and $d_{\mathrm{g}} = \lambda/10$. Since the current position is included in the candidates, $F(\hat{\mathbf{t}}_n) \geq F(\mathbf{t}_n)$. The spacing d_{g} is chosen as a fraction of the shortest period $\lambda/\max_{i,j\in\mathcal{M}\cup\{e\}}\|\mathbf{a}_j-\mathbf{a}_i\|$ of the phases in $F(\mathbf{t}_n)$, so that the basins of $F(\mathbf{t}_n)$ are densely sampled, and $\hat{\mathbf{t}}_n$ typically lies in the basin of a globally optimal solution of (P3.2.n).

3) Stage 2 (MM refinement): Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, we refine the position by the MM method [25], where $\mathbf{t}_n^{(k)}$ denotes the position of the n-th MA in the k-th MM iteration. To construct a surrogate function, we derive the gradient of $F(\mathbf{t}_n)$. For $u \in \{x_n, y_n\}$, let $\mathbf{D}_u \triangleq \mathrm{diag}(a_{1,u},\ldots,a_{M,u})$, where $a_{m,x} \triangleq \cos\theta_m\sin\phi_m$ and $a_{m,y} \triangleq \sin\theta_m$ are the entries of $\mathbf{a}_m$, and let $a_{e,u}$ denote the corresponding entry of $\mathbf{a}_e$. From (39)–(41), the derivatives of $\mathbf{l}_n$, $\mathbf{v}$, and $\mathbf{\Omega}$ w.r.t. u are given by

$$\dot{\mathbf{l}}_n \triangleq \frac{\partial \mathbf{l}_n}{\partial u} = -j\frac{2\pi}{\lambda}\mathbf{D}_u\mathbf{l}_n, \tag{44}$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u} = j\frac{2\pi}{\lambda}\big(a_{e,u}\mathbf{I}_M - \mathbf{D}_u\big)\mathbf{l}_n e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_e}, \tag{45}$$

$$\dot{\mathbf{\Omega}} \triangleq \frac{\partial \mathbf{\Omega}}{\partial u} = -\mathbf{\Omega}\big(\dot{\mathbf{l}}_n\mathbf{l}_n^H + \mathbf{l}_n\dot{\mathbf{l}}_n^H\big)\mathbf{\Omega}, \tag{46}$$

where (46) follows from $\partial\mathbf{\Phi}^{-1} = -\mathbf{\Phi}^{-1}(\partial\mathbf{\Phi})\mathbf{\Phi}^{-1}$. Based on (44)–(46), the derivatives $\dot{\omega}_m \triangleq \partial\omega_m/\partial u$, $\dot{c}_m \triangleq \partial c_m/\partial u$, and $\dot{d} \triangleq \partial d/\partial u$ are derived in closed form in Appendix B. Let $\chi_m \triangleq \xi_m p_m$ and $\Upsilon \triangleq q\psi + \bar{\sigma}_e^2$, whose derivatives are $\dot{\chi}_m = \kappa_e p_m \dot{c}_m$ and $\dot{\Upsilon} = \kappa_e q\dot{d}$, respectively. Moreover, $\dot{\eta}_m = -\eta_m\dot{\omega}_m/\omega_m$. By the chain rule,

$$\frac{\partial F}{\partial u} = \frac{1}{\ln 2}\sum_{m=1}^{M}\left[\frac{p_m\dot{\eta}_m}{1+\eta_m p_m} - \frac{\kappa_e p_m\dot{c}_m}{\chi_m+\Upsilon} + \frac{\kappa_e q\chi_m\dot{d}}{\Upsilon(\chi_m+\Upsilon)}\right], \tag{47}$$

and the gradient of F w.r.t. $\mathbf{t}_n$ is $\nabla_{\mathbf{t}_n}F = [\partial F/\partial x_n,\ \partial F/\partial y_n]^T$.

Since $F(\mathbf{t}_n)$ is twice continuously differentiable and $\mathcal{C}$ is compact, $\nabla_{\mathbf{t}_n}F$ is Lipschitz continuous over $\mathcal{C}$ with some constant $\mathcal{L}_n$. According to the descent lemma [26, Lemma 5.7], for any $\delta_n \geq \mathcal{L}_n$, the following quadratic function is a lower bound of $F(\mathbf{t}_n)$, which is tight at $\mathbf{t}_n^{(k)}$:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \triangleq F(\mathbf{t}_n^{(k)}) + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})^T\big(\mathbf{t}_n-\mathbf{t}_n^{(k)}\big) - \frac{\delta_n}{2}\big\|\mathbf{t}_n-\mathbf{t}_n^{(k)}\big\|^2. \tag{48}$$

Since $\mathcal{L}_n$ is difficult to obtain, δ_n is determined by backtracking [26], i.e., starting from $\max\{\delta_0, \delta_n^{\mathrm{pre}}/2\}$, where $\delta_0 > 0$ is a constant and δ_n^{pre} is the previously accepted value, δ_n is doubled until the update below satisfies $F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)})$, which holds once $\delta_n \geq \mathcal{L}_n$.

To handle the minimum-distance constraints, we note that $\|\mathbf{t}_n-\mathbf{t}_i\|^2$ is convex w.r.t. $\mathbf{t}_n$ and is thus lower-bounded by its first-order Taylor expansion at $\mathbf{t}_n^{(k)}$ [16], i.e.,

$$\|\mathbf{t}_n-\mathbf{t}_i\|^2 \geq \|\mathbf{\Delta}_{n,i}^{(k)}\|^2 + 2\big(\mathbf{\Delta}_{n,i}^{(k)}\big)^T\big(\mathbf{t}_n-\mathbf{t}_n^{(k)}\big), \tag{49}$$

where $\mathbf{\Delta}_{n,i}^{(k)} \triangleq \mathbf{t}_n^{(k)} - \mathbf{t}_i$. Replacing $\|\mathbf{t}_n-\mathbf{t}_i\|^2$ in the minimum-distance constraints with the right-hand side of (49) yields the linear constraint

$$\big(\mathbf{\Delta}_{n,i}^{(k)}\big)^T\mathbf{t}_n \geq \big(\mathbf{\Delta}_{n,i}^{(k)}\big)^T\mathbf{t}_n^{(k)} + \frac{D_{\min}^2 - \|\mathbf{\Delta}_{n,i}^{(k)}\|^2}{2}, \quad \forall i \neq n, \tag{50}$$

whose feasible set is a convex subset of that of the minimum-distance constraints w.r.t. $\mathbf{t}_n$. Based on (48) and (50), $\mathbf{t}_n$ is updated by solving

$$\mathbf{t}_n^{(k+1)} = \arg\max_{\mathbf{t}_n}\ \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \quad \text{s.t. } \mathbf{t}_n \in \mathcal{C},\ (50). \tag{51}$$

By completing the square, (51) is equivalent to projecting $\mathbf{t}_n^{(k)} + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})/\delta_n$ onto the convex polygon defined by $\mathcal{C}$ and (50). Its optimal solution lies at the projected point itself (if feasible), on an edge, or at a vertex of the polygon, and can thus be obtained exactly by enumerating the projections onto the $N+3$ constraint lines and their pairwise intersections. Since $\mathbf{t}_n^{(k)}$ is feasible for (51), the update satisfies

$$F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)}) \geq \hat{F}_n(\mathbf{t}_n^{(k)}|\mathbf{t}_n^{(k)}) = F(\mathbf{t}_n^{(k)}), \tag{52}$$

and $\mathbf{t}_n^{(k+1)}$ is feasible for (P3.2.n) due to (49). The MM iterations are repeated until the fractional increase of F is below ϵ_{in} or I_{in} iterations are reached.

D. Overall Algorithm, Convergence, and Complexity

The overall algorithm is summarized in Algorithm 2, whose convergence, initialization, and complexity are discussed below.

Convergence: In each iteration of Algorithm 2, neither the global search in (43) nor the MM updates in (52) decrease the objective value of (P3), and the power allocation is updated only if the objective value does not decrease. Hence, the objective value is nondecreasing over the iterations. Moreover, it is upper-bounded, e.g., $F \leq \sum_{m=1}^{M}\log_2(1+\zeta_m(N-M)(\kappa_m+1)P_{\mathrm{tot}})$ according to (18) and $p_m \leq P_{\mathrm{tot}}$. Therefore, the objective value of Algorithm 2 is guaranteed to converge.

Initialization: Since (P3) is nonconvex, the solution obtained by Algorithm 2 may depend on the initial positions. Owing to the global search in Stage 1, each MA can move to any

Algorithm 2 Two-Timescale AO Algorithm for (P3)

Input: Statistical CSI, P_{tot}, $\mathcal{C}$, $D_{\min}$, d_{g}, δ_0, ϵ_{in}, I_{in}, ϵ, and $I_{\max}$.
Output: $\mathbf{t}^\star$, $\mathbf{p}^\star$, and $q^\star$.

1: Initialize a feasible $\mathbf{t}^{(0)}$, obtain $(\mathbf{p}^{(0)}, q^{(0)})$ by Algorithm 1, and set $\ell = 0$.
2: **repeat**
3: **for** $n = 1$ to N **do**
4: *Stage 1:* Compute $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, and obtain $\hat{\mathbf{t}}_n$ by (43).
5: *Stage 2:* Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, update $\mathbf{t}_n$ by (51) with backtracking on δ_n until the fractional increase of F is below ϵ_{in} or I_{in} iterations are reached.
6: **end for**
7: Obtain $(\mathbf{p}', q')$ by Algorithm 1 for the updated positions $\mathbf{t}^{(\ell+1)}$. Set $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}', q')$ if this does not decrease F, and $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}^{(\ell)}, q^{(\ell)})$ otherwise.
8: $\ell \leftarrow \ell + 1$.
9: **until** the fractional increase of F is below ϵ or $\ell = I_{\max}$

feasible grid point in $\mathcal{C}$ in every iteration, which reduces this dependence compared with purely local updates. To further improve the solution, Algorithm 2 can be executed from several initializations, e.g., the FPA array and random feasible positions, and the solution with the largest objective value is selected.

Complexity: In each iteration, forming $\mathbf{\Phi}$ and $\mathbf{\Omega}$ requires $\mathcal{O}(NM^2 + M^3)$ operations. For each MA, $\mathbf{\Gamma}_n^{-1}$ is obtained from $\mathbf{\Omega}$ by a rank-one update. Stage 1 requires $\mathcal{O}(|\mathcal{G}|(M^2 + N))$ operations, including the feasibility check of the grid points, where $|\mathcal{G}| = (\lfloor A/d_{\mathrm{g}} \rfloor + 1)^2$. In Stage 2, the gradient in (47) and each evaluation of F require $\mathcal{O}(M^2)$ operations based on (42), and solving (51) by enumeration requires $\mathcal{O}(N^3)$ operations. Algorithm 1 requires $\mathcal{O}(QI_{\mathrm{b}}M)$ operations, where I_{b} is the number of bisection iterations. Hence, the complexity of Algorithm 2 is $\mathcal{O}\big(I_{\mathrm{o}}\big(NM^2 + M^3 + N|\mathcal{G}|(M^2+N) + NI_{\mathrm{in}}J(M^2+N^3) + QI_{\mathrm{b}}M\big)\big)$, where I_{o} is the number of AO iterations and J is the average number of backtracking trials. In contrast to stochastic optimization [18], this complexity does not depend on the number of channel realizations. Moreover, since Algorithm 2 depends only on the slowly varying statistical CSI, it is executed only once per frame, whereas the ZF precoding and AN generation in each CI require only $\mathcal{O}(NM^2 + M^3)$ operations.

V. Simulation Results

A. Simulation Setup and Benchmarks

Unless otherwise specified, we set $N = 8$, $M = 3$, $A = 4\lambda$, $D_{\min} = \lambda/2$, $\kappa_m = \kappa_e = 10$ dB, $\beta_m = \beta_e = 1$, and $\sigma_m^2 = \sigma_e^2 = \sigma^2$, such that $P_{\mathrm{tot}}/\sigma^2$ is the transmit SNR, which is 20 dB by default. The elevation and azimuth AoDs of the LUs are uniformly distributed within $[-\pi/3, \pi/3]$. Similar to [15], Eve is located in a direction close to that of LU 1 to model the challenging case in Section I. Specifically, its AoDs are $\theta_e = \theta_1 + \Delta\cos\tilde{\phi}$ and $\phi_e = \phi_1 + \Delta\sin\tilde{\phi}$, i.e., they deviate from those of LU 1 by the angular separation Δ in the direction specified by $\tilde{\phi}$, which is uniformly distributed over $[0, 2\pi)$. By default, $\Delta = 0.1$ rad, for which the normalized LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|/N$ between Eve and LU 1 exceeds 0.94 for the FPA, i.e., Eve lies well within the main lobe of the beam steered toward LU 1. The case where the AoDs of Eve are independent of those of the LUs is considered in Section V-F.

For Algorithm 2, the grid spacing in Stage 1 is $d_{\mathrm{g}} = \lambda/10$, which is less than one third of the shortest phase period in Section IV-C. The MM refinement in Stage 2 uses $\delta_0 = 1$ and stops when the fractional increase of F is below $\epsilon_{\mathrm{in}} = 10^{-6}$ or after $I_{\mathrm{in}} = 20$ iterations, and Algorithm 2 stops when the fractional increase of F is below $\epsilon = 10^{-4}$ or after $I_{\max} = 50$ iterations. Algorithm 1 uses $Q = 41$ grid points and 60 bisection iterations. Algorithm 2 is executed from three initializations, i.e., a 2×4 uniform planar array (UPA) with $\lambda/2$ spacing and two random feasible positions, and the best solution is selected. All the results are averaged over 100 random realizations of the AoDs of the LUs and Eve shared by all the schemes, which is sufficient for the comparisons, since the standard error of the average ESSR of the proposed design at 20 dB is 0.11 bps/Hz, i.e., 0.5% of its value. Since the closed-form approximations are used only for optimization, the ergodic rates of each design are evaluated by averaging the instantaneous rates based on the exact SINRs in (12) and (13) over 5×10^3 channel realizations.

We compare the proposed design with three benchmarks: 1) **Rate-oriented MA**: The antenna positions maximize the approximate ergodic sum rate of the LUs with equal power, i.e., $\sum_{m=1}^{M} \log_2(1 + \eta_m(\mathbf{t})P_{\mathrm{tot}}/M)$, which ignores Eve [16]. For a fair comparison, this problem is solved by the position update in Section IV-C with the same initializations. This benchmark reveals the gain of the secrecy-aware positioning enabled by Proposition 1. 2) **AS**: Similar to [15], the BS selects N of $2N$ candidate antennas, which form a UPA with 4 rows and 4 columns and $\lambda/2$ spacing, by an exhaustive search over all the 12870 possible subsets, where the selection metric is the approximate ESSR in (P3). AS requires no antenna movement but twice as many antennas and an RF switching network. 3) **FPA**: The BS adopts the 2×4 UPA with $\lambda/2$ spacing. All the schemes adopt the power allocation in Algorithm 1, so that their performance differences are caused only by the antenna positions.

B. Accuracy of the Closed-Form Approximations

Fig. 3 evaluates the accuracy of Lemma 1 and Proposition 1 at the designs obtained by Algorithm 2, i.e., at the operating points selected by the optimization, where 10^4 channel realizations are used for the Monte Carlo simulations. It is observed that Proposition 1 closely matches the Monte Carlo results over the entire ranges of the SNR and the Rician factor, with a maximum absolute error of 0.05 bps/Hz. In contrast, Lemma 1 underestimates the ergodic sum rate of the LUs, and the gap increases with the Rician factor and saturates at high SNR. The gap is caused by Jensen's inequality and the central Wishart approximation in Lemma 1, which is exact for Rayleigh fading but overestimates $\mathbb{E}\{[(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}\}$ by a factor approaching $N/(N-M)$ as the LoS components dominate, which leads to a gap of up to about $M\log_2(N/(N-M)) \approx 2$ bps/Hz in the sum rate for $N = 8$ and $M = 3$. Hence, the approximate ESSR in (25) is a conservative estimate of the actual ESSR. Nevertheless, at each SNR in Fig. 5(a), this gap

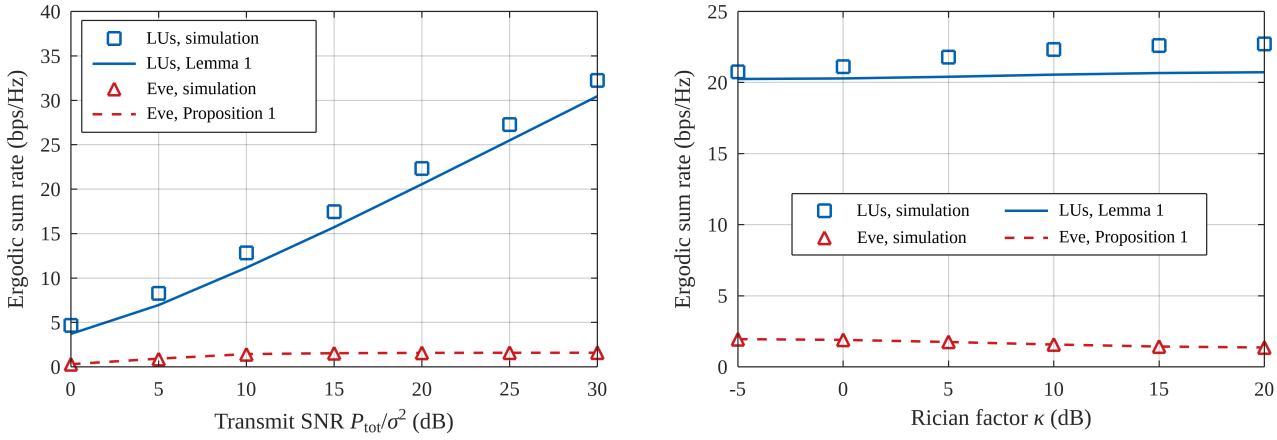


Fig. 3. Ergodic sum rates of the LUs and Eve achieved by the proposed design, obtained by Monte Carlo simulations (markers) and by Lemma 1 and Proposition 1 (lines), versus (a) the transmit SNR P_{tot}/σ^2 and (b) the Rician factor $\kappa = \kappa_m = \kappa_e$.

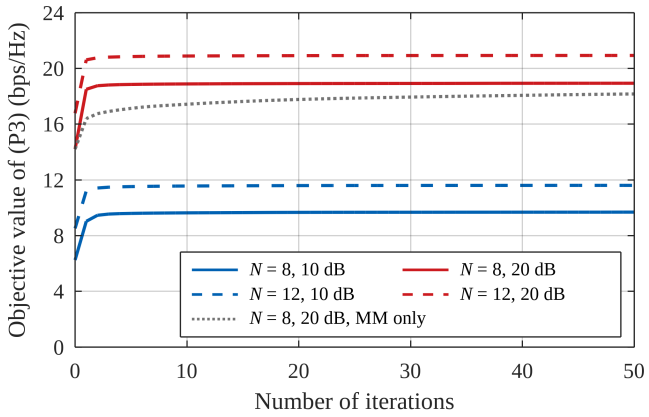


Fig. 4. Convergence behaviour of Algorithm 2 for different numbers of MAs N and transmit SNRs P_{tot}/σ^2, averaged over 100 realizations of the AoDs (UPA initialization). "MM only": Algorithm 2 without the global search in Stage 1.

differs by less than 0.3 bps/Hz among the four schemes, which is smaller than the ESSR gains of the proposed design. Hence, the approximation preserves the advantage of the proposed design over the benchmarks and is suitable as the design objective.

C. Convergence Behaviour of Algorithm 2

Fig. 4 shows the objective value of (P3) versus the number of iterations of Algorithm 2 for $N \in \{8, 12\}$ and $P_{\text{tot}}/\sigma^2 \in \{10, 20\}$ dB, where the UPA initialization is adopted (3×4 UPA for $N = 12$). To show the complete trajectories, Algorithm 2 is run for 50 iterations without the stopping criterion. It is observed that, for all the settings, the objective value increases monotonically and converges within about 15 iterations, where most of the improvement is achieved in the first few iterations. The fast convergence is due to the global search in Stage 1, which moves each MA directly to the best basin of $F(\mathbf{t}_n)$, so that the positions do not need to traverse the movable region through local updates. In contrast, with only the MM updates, the objective value increases much more slowly and remains lower after 50 iterations, since the MAs are trapped in the local maxima near their initial positions.

D. ESSR and AN Power Versus Transmit SNR

Fig. 5(a) shows that the proposed design achieves the highest ESSR over the entire SNR range. At $P_{\text{tot}}/\sigma^2 = 20$ dB, it improves the ESSR by 14.8%, 20.4%, and 31.2% compared with the rate-oriented MA, AS, and the FPA, respectively. Although the rate-oriented MA exploits the same movable region, its gain over the FPA is less than half of that of the proposed design. The reason is that the rate-oriented positions only decorrelate the LUs, whereas the proposed positions additionally reduce the information leakage and increase the AN received by Eve (see Section III-D). Moreover, AS improves the ESSR over the FPA by only 9.0%, since its $2N$ candidates form a compact UPA with an aperture of only $1.5\lambda \times 1.5\lambda$, which provides limited spatial resolution for decorrelating the LoS components of Eve and LU 1. As the SNR increases, the ESSR gains of the proposed design approach constants, since the antenna positions improve the constant term rather than the slope of the ESSR at high SNR (see Section III-D).

Fig. 5(b) shows that the optimized AN power fraction increases with the SNR and saturates at high SNR, as predicted by (29). Moreover, the proposed design requires considerably less AN power than all the benchmarks, whereas the rate-oriented MA saves little AN power compared with the FPA. This result verifies the AN-saving effect in Section III-D: since the proposed positions reduce the information leakage and increase the AN received by Eve, more power can be allocated to the confidential signals.

E. Robustness to the Eavesdropper Model and the AoD Error of Eve

Tables I and II examine whether the performance of the designs relies on the worst-case eavesdropper model in (6)

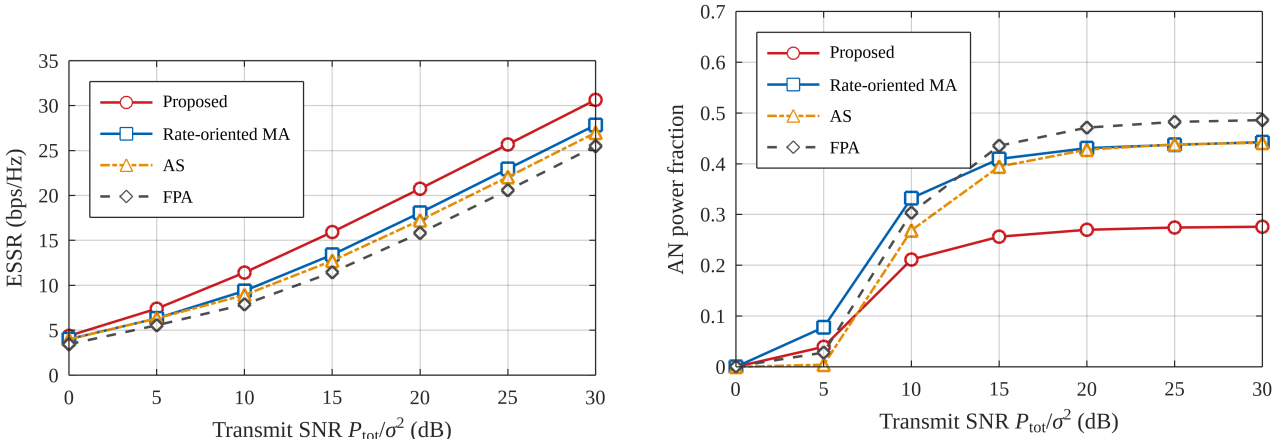


Fig. 5. Performance versus the transmit SNR $P_{\rm tot}/\sigma^2$: (a) ESSR; (b) optimized AN power fraction $(N-M)q/P_{\rm tot}$.

TABLE I
ESSR (BPS/HZ) UNDER DIFFERENT EAVESDROPPER MODELS

	Worst-case Eve (6)			TIN Eve (53)		
$P_{\rm tot}/\sigma^2$ (dB)	10	20	30	10	20	30
Proposed	**11.42**	**20.75**	**30.64**	**11.85**	**21.24**	**31.14**
Rate-oriented MA	9.38	18.08	27.84	10.53	19.55	29.38
AS	8.92	17.24	26.98	9.77	18.58	28.39
FPA	7.88	15.82	25.49	8.70	17.24	27.00
Proposed w/o AN	9.94	13.06	13.85	11.00	18.70	27.14

TABLE II
ESSR (BPS/HZ) VERSUS THE AOD ERROR $\varepsilon_{\rm e}$ OF EVE ($P_{\rm tot}/\sigma^2 = 20$ DB)

$\varepsilon_{\rm e}$ (rad)	0	0.01	0.02	0.05	0.1
Proposed	**20.75**	**20.73**	**20.65**	**20.13**	**18.73**
Rate-oriented MA	18.08	18.07	18.07	18.02	17.93
AS	17.24	17.24	17.22	17.16	16.97
FPA	15.82	15.82	15.81	15.81	15.80

and on the known statistical CSI of Eve, respectively. Table I evaluates the designs also against an Eve that treats the signals of the other LUs as interference (TIN), whose SINR is

$$\gamma_{e,m}^{\rm TIN} = \frac{p_m|\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2}{\sum_{i=1,i\neq m}^{M} p_i|\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_i|^2 + q\,\mathbf{g}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t}) + \sigma_e^2}. \tag{53}$$

Two observations can be made. First, the proposed design achieves the highest ESSR under both models, which confirms that its gain is not an artifact of the worst-case assumption. Moreover, its ESSR under the worst-case model is only slightly lower than that against the TIN Eve, i.e., the secrecy guarantee of the worst-case model comes at a marginal cost. Second, for the proposed design without AN, i.e., Algorithm 2 with $q = 0$, the ESSR increases with the SNR against the TIN Eve, for which the inter-user interference acts as AN, but saturates under the worst-case model [4]. Hence, the AN is indispensable for a secrecy performance that does not rely on the decoding capability of Eve.

Table II considers imperfect statistical CSI of Eve, where all the schemes are designed with an AoD of Eve that deviates from the true one by $\varepsilon_{\rm e}$ in a random direction, and are evaluated with the true AoD. It is observed that the ESSR of the proposed design decreases only slightly for $\varepsilon_{\rm e} \leq 0.05$ rad, and the proposed design still outperforms all the benchmarks for $\varepsilon_{\rm e} = \Delta = 0.1$ rad, i.e., an error as large as the angular separation between Eve and LU 1. This robustness is due to the smooth variation of the LoS correlations between Eve and the LUs with the AoD of Eve, so that the positions optimized for a moderately erroneous AoD still decorrelate the actual LoS components.

F. Impact of the Region Size and the Angular Separation

Fig. 6(a) shows the ESSR versus the side length A of the movable region, where we also consider AS with region-wide candidates, i.e., a 4×4 candidate UPA spanning $\mathcal{C}$ with spacing $A/3$. Note that the FPA and AS with the compact candidate array do not depend on A. It is observed that the gain of the proposed design over the FPA increases from 9.2% to 38.9% as A increases from 1.5λ to 6λ. The reason is that a larger region provides more spatial degrees of freedom for decorrelating the LoS components of Eve and the LUs. In contrast, the ESSR of the rate-oriented MA increases much more slowly with A, since it does not exploit the enlarged region to reduce the information leakage. AS with region-wide candidates also benefits from the enlarged aperture and the Eve-aware selection metric, but the proposed design still outperforms it for $A \geq 2\lambda$ with half the number of antennas, since the MAs can be placed at arbitrary positions in $\mathcal{C}$ rather than at $2N$ fixed candidate positions. For $A = 1.5\lambda$, the two AS schemes coincide and perform comparably to the proposed design, since the candidate array covers the entire region.

Fig. 6(b) shows the ESSR versus the angular separation Δ for $M = 3$ and $M = 1$. For $M = 3$, all the schemes benefit from a larger Δ, since Eve becomes more separable from LU 1. For $\Delta = 0.02$ rad, Eve is almost aligned with LU 1, and the proposed design outperforms the rate-oriented MA only slightly, since their LoS components can hardly be decorrelated with an aperture of 4λ. As Δ increases, the gain of the proposed design first increases and then decreases, since

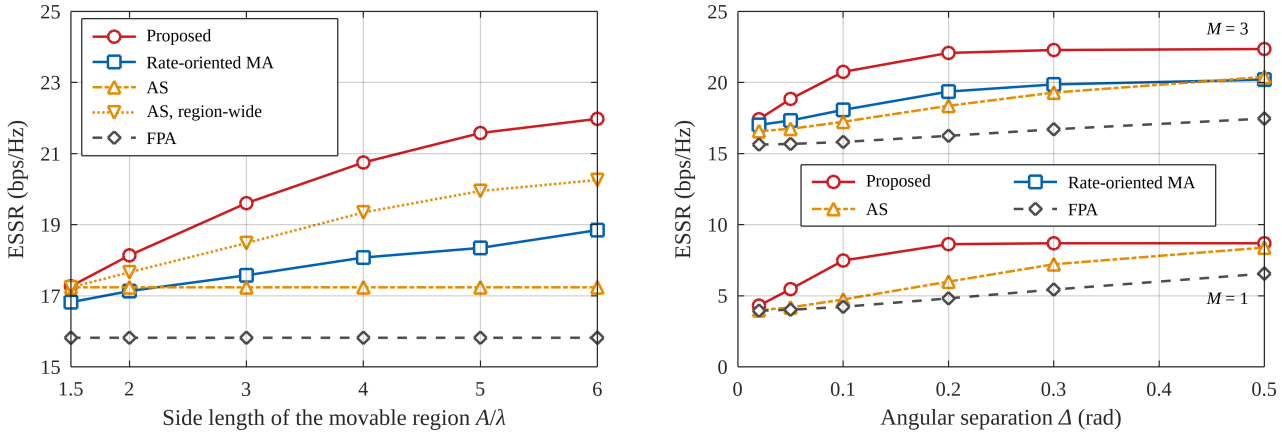


Fig. 6. ESSR versus (a) the side length A of the movable region and (b) the angular separation Δ between Eve and LU 1 for $M = 3$ and $M = 1$.

Eve eventually becomes separable even with Eve-unaware positions. For $M = 1$, the rate-oriented MA coincides with the FPA, since the ergodic rate of the LU does not depend on the antenna positions (see Section III-D). In contrast, the proposed design significantly improves the ergodic secrecy rate, e.g., by 77.1% over the FPA for $\Delta = 0.1$ rad, by reducing the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ in (26) and (27). This result confirms that, different from the rate-oriented design, the MAs enhance secrecy even for a single LU. Finally, when the AoDs of Eve are independent of those of the LUs, i.e., uniformly distributed within $[-\pi/3, \pi/3]$, the proposed design improves the ESSR by 10.3% and 21.8% over the rate-oriented MA and the FPA, respectively, for $M = 3$, and the secrecy rate by 13.1% and 2.1% over the FPA and AS, respectively, for $M = 1$. Hence, the secrecy gain of the MAs is most pronounced when Eve is close to an LU, but does not rely on this placement.

G. Benefit of the Long-Term Power Allocation

Fig. 7 evaluates the benefit of the proposed long-term power allocation, i.e., the secrecy water-filling in Proposition 2 combined with the optimized AN power. It is compared with two EP benchmarks with $p_m = \alpha P_{\text{tot}}/M$ and $q = (1-\alpha)P_{\text{tot}}/(N-M)$, where α is either optimized by a one-dimensional search, similar to [5], or fixed to 0.5, similar to [15]. For all the schemes, the antenna positions are optimized by Algorithm 2, where the objective of the EP benchmarks retains $[\cdot]^+$, since the LUs cannot be switched off (see Section IV-A). The large-scale fading of the LUs is $[\beta_1, \beta_2, \beta_3] = [0, -\varsigma/2, -\varsigma]$ dB, where ς is referred to as the fading spread.

Fig. 7(a) shows that, for $\varsigma = 20$ dB, the proposed power allocation outperforms both EP benchmarks. Its gain over EP with optimized α is significant at low-to-moderate SNR, e.g., 0.65 bps/Hz at 10 dB, but becomes marginal at 20 dB. The reason is that the proposed power allocation switches off or reduces the power of the LUs whose effective channel gains are comparable to that of Eve (Proposition 2), whereas at high SNR, all the LUs achieve positive secrecy rates and nearly equal powers become optimal, similar to conventional water-filling. Fig. 7(b) shows that EP with optimized α performs comparably to the proposed power allocation for small fading spreads, e.g., $\varsigma \leq 5$ dB, since the LUs have similar effective channel gains, which are further balanced by the optimized positions, whereas the gain of the proposed power allocation increases with ς for larger fading spreads. Moreover, EP with $\alpha = 0.5$ performs worst in all the cases, especially at low SNR, where little or no AN is needed. Hence, both components of the long-term power allocation are important: the information–AN power splitting improves the ESSR in all the cases, whereas the per-user power allocation provides additional gains for heterogeneous LUs at low-to-moderate SNR.

VI. Conclusion

This paper investigated two-timescale secure transmission for an MA-enabled multiuser downlink system with a passive Eve. The derived closed-form approximation of the ergodic wiretap rate revealed that, beyond decorrelating the LUs, the MAs enhance secrecy by reducing the information leakage and increasing the AN received by Eve. Based on this result, we developed an AO algorithm, which combines secrecy water-filling with a two-stage position update and exhibits fast and monotonic convergence. Simulation results demonstrated that the proposed secrecy-aware positioning significantly outperforms the rate-oriented two-timescale MA design, AS, and FPA arrays, enhances the secrecy even for a single LU, and reduces the AN power required for secure transmission. Since this paper assumes a single-antenna Eve and perfect CSI of the LUs, extending the design to multiple multi-antenna eavesdroppers and imperfect CSI is an important direction for future work.

Appendix A
Proof of Proposition 1

We first establish the following lemma for a generic non-central Gaussian matrix, and then apply it to (21).

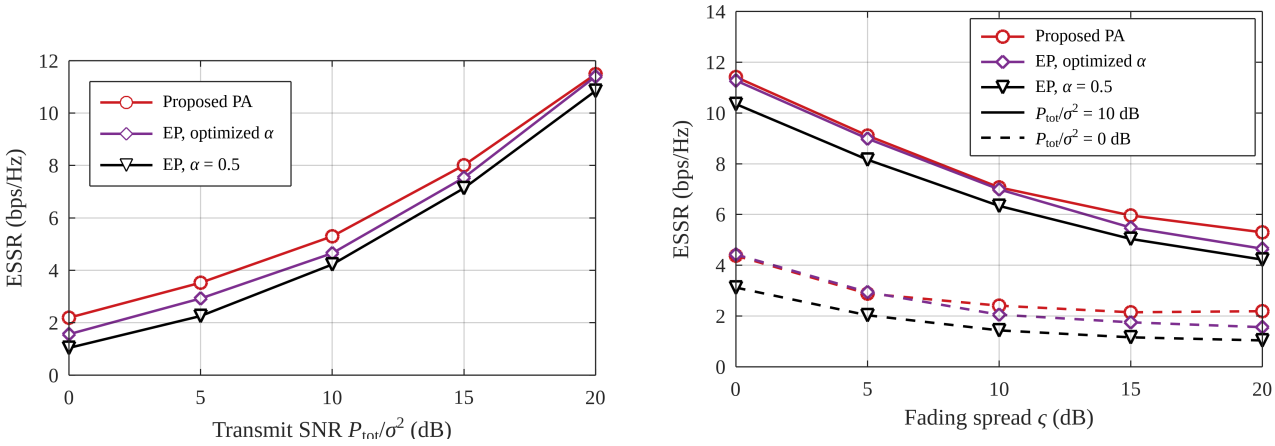


Fig. 7. ESSR of the proposed power allocation (PA) and two equal-power (EP) benchmarks versus (a) the transmit SNR with $\varsigma = 20$ dB and (b) the fading spread ς.

Lemma 3: Let $\mathbf{F} = \bar{\mathbf{F}} + \mathbf{W} \in \mathbb{C}^{N\times k}$ with $k < N$, where $\bar{\mathbf{F}}$ is deterministic and the entries of $\mathbf{W}$ are i.i.d. $\mathcal{CN}(0,1)$ random variables. Let $\mathbf{b} \in \mathbb{C}^{N\times 1}$ be a deterministic vector, and define $\mathbf{\Phi_F} \triangleq \mathbb{E}\{\mathbf{F}^H\mathbf{F}\} = \bar{\mathbf{F}}^H\bar{\mathbf{F}} + N\mathbf{I}_k$, $\bar{\mathbf{w}}_m(\mathbf{F}) \triangleq \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{e}_m/\|\mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{e}_m\|$, and $\mathbf{P}_{\mathbf{F}}^{\perp} \triangleq \mathbf{I}_N - \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{F}^H$. Then, we have

$$\mathbb{E}\{|\mathbf{b}^H\bar{\mathbf{w}}_m(\mathbf{F})|^2\} \approx \frac{\left|[\mathbf{\Phi}_{\mathbf{F}}^{-1}\bar{\mathbf{F}}^H\mathbf{b}]_m\right|^2 + \|\mathbf{b}\|^2[\mathbf{\Phi}_{\mathbf{F}}^{-2}]_{m,m}}{[\mathbf{\Phi}_{\mathbf{F}}^{-1}]_{m,m}}, \tag{54}$$

$$\mathbb{E}\{\mathbf{b}^H\mathbf{P}_{\mathbf{F}}^{\perp}\mathbf{b}\} \approx \frac{(N-k)\left(\|\mathbf{b}\|^2 - \mathbf{b}^H\bar{\mathbf{F}}\mathbf{\Phi}_{\mathbf{F}}^{-1}\bar{\mathbf{F}}^H\mathbf{b}\right)}{N-k+N\mathrm{tr}(\mathbf{\Phi}_{\mathbf{F}}^{-1})}. \tag{55}$$

Moreover, (54) and (55) are exact for $\bar{\mathbf{F}} = \mathbf{0}$, and are asymptotically exact for $\bar{\mathbf{F}} = \varrho\bar{\mathbf{F}}_0$ as $\varrho \to \infty$, where $\bar{\mathbf{F}}_0$ has full column rank. The right-hand sides of (54) and (55) lie in $[0, \|\mathbf{b}\|^2]$, as do their left-hand sides.

Proof: For brevity, let $\mathbf{\Phi} \triangleq \mathbf{\Phi_F}$. We first prove (54). Replacing $(\mathbf{F}^H\mathbf{F})^{-1}$ in the unnormalized ZF direction by $\mathbf{\Phi}^{-1} = (\mathbb{E}\{\mathbf{F}^H\mathbf{F}\})^{-1}$ yields $\bar{\mathbf{w}}_m(\mathbf{F}) \approx \mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m/\|\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m\|$. Then, we approximate the mean of $|\mathbf{b}^H\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m|^2/\|\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m\|^2$ by the ratio of the means of its numerator and denominator, in the same spirit as [19, Lemma 1]. Since $\mathbb{E}\{\mathbf{W}\} = \mathbf{0}$ and $\mathbb{E}\{\mathbf{W}\mathbf{x}\mathbf{x}^H\mathbf{W}^H\} = \|\mathbf{x}\|^2\mathbf{I}_N$ for any deterministic $\mathbf{x}$, we have

$$\begin{aligned}\mathbb{E}\{|\mathbf{b}^H\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m|^2\} &= |\mathbf{b}^H\bar{\mathbf{F}}\mathbf{\Phi}^{-1}\mathbf{e}_m|^2 + \|\mathbf{b}\|^2\|\mathbf{\Phi}^{-1}\mathbf{e}_m\|^2 \\ &= \left|[\mathbf{\Phi}^{-1}\bar{\mathbf{F}}^H\mathbf{b}]_m\right|^2 + \|\mathbf{b}\|^2[\mathbf{\Phi}^{-2}]_{m,m},\end{aligned} \tag{56}$$

$$\mathbb{E}\{\|\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m\|^2\} = \mathbf{e}_m^T\mathbf{\Phi}^{-1}\mathbf{\Phi}\mathbf{\Phi}^{-1}\mathbf{e}_m = [\mathbf{\Phi}^{-1}]_{m,m}, \tag{57}$$

which lead to (54). Next, we prove (55). Replacing $(\mathbf{F}^H\mathbf{F})^{-1}$ in $\mathbf{P_F} \triangleq \mathbf{I}_N - \mathbf{P}_{\mathbf{F}}^{\perp} = \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{F}^H$ by $\mathbf{\Phi}^{-1}$ and using $\mathbb{E}\{\mathbf{W}\mathbf{\Phi}^{-1}\mathbf{W}^H\} = \mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N$ yield

$$\mathbb{E}\{\mathbf{P_F}\} \approx \mathbf{\Pi} + \mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N, \qquad \mathbf{\Pi} \triangleq \bar{\mathbf{F}}\mathbf{\Phi}^{-1}\bar{\mathbf{F}}^H, \tag{58}$$

where $\mathbf{\Pi}$ and $\mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N$ represent the parts of the signal subspace induced by the LoS and NLoS components, respectively. However, (58) lets the NLoS component capture the fraction $\mathrm{tr}(\mathbf{\Phi}^{-1})$ of every direction, including those already captured by the LoS component, so that $\mathbf{b}^H(\mathbf{I}_N - \mathbf{\Pi} - \mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N)\mathbf{b}$ may even become negative when $\mathbf{b}$ is closely aligned with the column space of $\bar{\mathbf{F}}$. To avoid this double counting, we let the NLoS component capture a fraction $\vartheta \in [0,1]$ of the complement of $\mathbf{\Pi}$ only, i.e.,

$$\mathbb{E}\{\mathbf{P_F}\} \approx \mathbf{\Pi} + \vartheta(\mathbf{I}_N - \mathbf{\Pi}), \tag{59}$$

which satisfies $\mathbf{0} \preceq \mathbf{\Pi} + \vartheta(\mathbf{I}_N - \mathbf{\Pi}) \preceq \mathbf{I}_N$ since $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$, as shown below. The fraction ϑ is determined by the identity $\mathrm{tr}(\mathbf{P_F}) = k$, which holds for every realization of $\mathbf{F}$. Since $\mathrm{tr}(\mathbf{\Pi}) = \mathrm{tr}\big(\mathbf{\Phi}^{-1}(\mathbf{\Phi} - N\mathbf{I}_k)\big) = k - N\mathrm{tr}(\mathbf{\Phi}^{-1})$, taking the trace of both sides of (59) yields $k - N\mathrm{tr}(\mathbf{\Phi}^{-1}) + \vartheta\big(N - k + N\mathrm{tr}(\mathbf{\Phi}^{-1})\big) = k$, i.e., $\vartheta = N\mathrm{tr}(\mathbf{\Phi}^{-1})/(N - k + N\mathrm{tr}(\mathbf{\Phi}^{-1}))$. Therefore, $\mathbb{E}\{\mathbf{P}_{\mathbf{F}}^{\perp}\} \approx (1-\vartheta)(\mathbf{I}_N - \mathbf{\Pi})$ with $1 - \vartheta = (N-k)/(N - k + N\mathrm{tr}(\mathbf{\Phi}^{-1}))$, which leads to (55).

Range: The nonzero eigenvalues of $\mathbf{\Pi}$ are $\mu_i^2/(\mu_i^2+N) < 1$, where $\mu_1, \ldots, \mu_k$ are the singular values of $\bar{\mathbf{F}}$, so that $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$. Moreover, $\mathbf{\Phi} \succeq N\mathbf{I}_k$ implies $0 < \mathrm{tr}(\mathbf{\Phi}^{-1}) \leq k/N$, i.e., $1 - \vartheta \in [1 - k/N, 1)$. Hence, the right-hand side of (55) lies in $[0, \|\mathbf{b}\|^2]$. For (54), the Cauchy–Schwarz inequality and $N \geq 1$ imply that the numerator is at most $\|\mathbf{b}\|^2\mathbf{e}_m^T\mathbf{\Phi}^{-1}(\bar{\mathbf{F}}^H\bar{\mathbf{F}} + N\mathbf{I}_k)\mathbf{\Phi}^{-1}\mathbf{e}_m = \|\mathbf{b}\|^2[\mathbf{\Phi}^{-1}]_{m,m}$, so that the right-hand side of (54) lies in $[0, \|\mathbf{b}\|^2]$.

Exactness for $\bar{\mathbf{F}} = \mathbf{0}$: In this case, $\mathbf{\Phi} = N\mathbf{I}_k$ and $\mathbf{\Pi} = \mathbf{0}$, and the right-hand sides of (54) and (55) reduce to $\|\mathbf{b}\|^2/N$ and $\|\mathbf{b}\|^2(1 - k/N)$, respectively. Since $\mathbf{QW}$ has the same distribution as $\mathbf{W}$ for any unitary $\mathbf{Q}$, and $\bar{\mathbf{w}}_m(\mathbf{QW}) = \mathbf{Q}\bar{\mathbf{w}}_m(\mathbf{W})$, $\bar{\mathbf{w}}_m(\mathbf{W})$ is uniformly distributed on the unit sphere, and $\mathbb{E}\{\mathbf{P}_{\mathbf{W}}^{\perp}\}$ commutes with every unitary matrix and thus equals $(1-k/N)\mathbf{I}_N$ due to $\mathrm{tr}(\mathbf{P}_{\mathbf{W}}^{\perp}) = N - k$. Hence, the left-hand sides coincide with the right-hand sides.

Asymptotic exactness for $\bar{\mathbf{F}} = \varrho\bar{\mathbf{F}}_0$: As $\varrho \to \infty$, $\mathbf{F}/\varrho \to \bar{\mathbf{F}}_0$ almost surely. Since $\bar{\mathbf{w}}_m(\cdot)$ and $\mathbf{P}_{(\cdot)}^{\perp}$ are invariant to the scaling of their arguments and continuous at full-column-rank matrices, and $|\mathbf{b}^H\bar{\mathbf{w}}_m(\mathbf{F})|^2$ and $\mathbf{b}^H\mathbf{P}_{\mathbf{F}}^{\perp}\mathbf{b}$ are bounded by $\|\mathbf{b}\|^2$, the dominated convergence theorem yields $\mathbb{E}\{|\mathbf{b}^H\bar{\mathbf{w}}_m(\mathbf{F})|^2\} \to |\mathbf{b}^H\bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$ and $\mathbb{E}\{\mathbf{b}^H\mathbf{P}_{\mathbf{F}}^{\perp}\mathbf{b}\} \to$

$\mathbf{b}^H\mathbf{P}_{\bar{\mathbf{F}}_0}^{\perp}\mathbf{b}$. On the other hand, $\varrho^2\mathbf{\Phi}^{-1} \to (\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}$, $\mathbf{\Pi} \to \bar{\mathbf{F}}_0(\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}\bar{\mathbf{F}}_0^H$, and $\mathrm{tr}(\mathbf{\Phi}^{-1}) \to 0$. Hence, the right-hand side of (55) converges to $\mathbf{b}^H\mathbf{P}_{\bar{\mathbf{F}}_0}^{\perp}\mathbf{b}$, and that of (54) converges to $|[(\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}\bar{\mathbf{F}}_0^H\mathbf{b}]_m|^2/[(\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}]_{m,m} = |\mathbf{b}^H\bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$, since the second term in the numerator of (54) decays as $\mathcal{O}(\varrho^{-4})$ whereas the denominator decays as $\mathcal{O}(\varrho^{-2})$. This completes the proof of Lemma 3. ∎

Proof of Proposition 1: From (15), $\bar{\mathbf{w}}_m = \bar{\mathbf{w}}_m(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})$ and $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{P}^{\perp}_{\mathbf{L}(\mathbf{t})+\tilde{\mathbf{H}}}$. Applying Lemma 3 with $\bar{\mathbf{F}} = \mathbf{L}(\mathbf{t})$, $\mathbf{W} = \tilde{\mathbf{H}}$, $k = M$, and $\mathbf{b} = \bar{\mathbf{g}}(\mathbf{t})$, for which $\mathbf{\Phi}_{\mathbf{F}} = \mathbf{\Phi}(\mathbf{t})$, $\bar{\mathbf{F}}^H\mathbf{b} = \mathbf{v}(\mathbf{t})$, and $\|\mathbf{b}\|^2 = N$, yields $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{y\} \approx d(\mathbf{t})$. Substituting them into (21) yields (24), which completes the proof.

APPENDIX B
DERIVATIVES OF ω_m, c_m, AND d

Let $o_m \triangleq [\mathbf{\Omega}^2]_{m,m}$, $\tau \triangleq \mathrm{tr}(\mathbf{\Omega})$, $\tilde{\mathbf{v}} \triangleq \mathbf{\Omega}\mathbf{v}$, and $\varpi \triangleq \mathbf{v}^H\mathbf{\Omega}\mathbf{v}$. Since $\mathbf{\Omega}$ is Hermitian, (46) can be rewritten as $\dot{\mathbf{\Omega}} = -(\mathbf{\Omega}\dot{\mathbf{l}}_n)(\mathbf{\Omega}\mathbf{l}_n)^H - (\mathbf{\Omega}\mathbf{l}_n)(\mathbf{\Omega}\dot{\mathbf{l}}_n)^H$. Hence, the derivatives of ω_m, o_m, and τ w.r.t. $u \in \{x_n, y_n\}$ are given by

$$\dot{\omega}_m = [\dot{\mathbf{\Omega}}]_{m,m} = -2\mathrm{Re}\big\{[\mathbf{\Omega}\dot{\mathbf{l}}_n]_m[\mathbf{\Omega}\mathbf{l}_n]_m^*\big\}, \tag{60}$$

$$\dot{o}_m = 2\mathrm{Re}\big\{[\mathbf{\Omega}\dot{\mathbf{\Omega}}]_{m,m}\big\} = -2\mathrm{Re}\big\{[\mathbf{\Omega}^2\dot{\mathbf{l}}_n]_m[\mathbf{\Omega}\mathbf{l}_n]_m^* + [\mathbf{\Omega}^2\mathbf{l}_n]_m[\mathbf{\Omega}\dot{\mathbf{l}}_n]_m^*\big\}, \tag{61}$$

$$\dot{\tau} = \mathrm{tr}(\dot{\mathbf{\Omega}}) = -2\mathrm{Re}\big\{\mathbf{l}_n^H\mathbf{\Omega}^2\dot{\mathbf{l}}_n\big\}, \tag{62}$$

where (61) follows from $[\dot{\mathbf{\Omega}}\mathbf{\Omega}]_{m,m} = [\mathbf{\Omega}\dot{\mathbf{\Omega}}]_{m,m}^*$. For c_m in (22), the derivative of $\tilde{\mathbf{v}} = \mathbf{\Omega}\mathbf{v}$ is

$$\dot{\tilde{\mathbf{v}}} = \dot{\mathbf{\Omega}}\mathbf{v} + \mathbf{\Omega}\dot{\mathbf{v}} = -\mathbf{\Omega}\dot{\mathbf{l}}_n\big(\mathbf{l}_n^H\tilde{\mathbf{v}}\big) - \mathbf{\Omega}\mathbf{l}_n\big(\dot{\mathbf{l}}_n^H\tilde{\mathbf{v}}\big) + \mathbf{\Omega}\dot{\mathbf{v}}, \tag{63}$$

and thus

$$\dot{c}_m = \frac{2\mathrm{Re}\big\{[\tilde{\mathbf{v}}]_m^*[\dot{\tilde{\mathbf{v}}}]_m\big\} + N\dot{o}_m - c_m\dot{\omega}_m}{\omega_m}. \tag{64}$$

For $d = (1-\vartheta)(N-\varpi)$ in (23), where $1-\vartheta = (N-M)/(N-M+N\tau)$ as in Appendix A, since $\mathbf{v}^H\dot{\mathbf{\Omega}}\mathbf{v} = -2\mathrm{Re}\{(\tilde{\mathbf{v}}^H\dot{\mathbf{l}}_n)(\mathbf{l}_n^H\tilde{\mathbf{v}})\}$, we have

$$\dot{\varpi} = 2\mathrm{Re}\{\tilde{\mathbf{v}}^H\dot{\mathbf{v}}\} - 2\mathrm{Re}\big\{(\tilde{\mathbf{v}}^H\dot{\mathbf{l}}_n)(\mathbf{l}_n^H\tilde{\mathbf{v}})\big\}, \tag{65}$$

$$\dot{d} = -(1-\vartheta)\left(\dot{\varpi} + \frac{N(N-\varpi)\,\dot{\tau}}{N-M+N\tau}\right). \tag{66}$$

Substituting $\dot{\eta}_m = -\eta_m\dot{\omega}_m/\omega_m$, (64), and (66) into (47) yields $\nabla_{\mathbf{t}_n}F$. All the quantities in (60)–(66) are obtained from matrix-vector products with $\mathbf{\Omega}$, which require $\mathcal{O}(M^2)$ operations.

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